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(54) **IMPROVED PROTEIN PURIFICATION**

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(57) **ABSTRACT**

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The present invention relates to protein purification, primarily in the chromatographic field. More closely, the invention relates to affinity chromatography using a split intein system comprising a C-intein tag and N-intein ligand, wherein the N-intein ligand provides increased solubility suitable for large scale purification of any recombinant target protein.

(30) **Foreign Application Priority Data**

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Specification includes a Sequence Listing.

% of SDS according to Biacore, CFCA using different extraction-methods

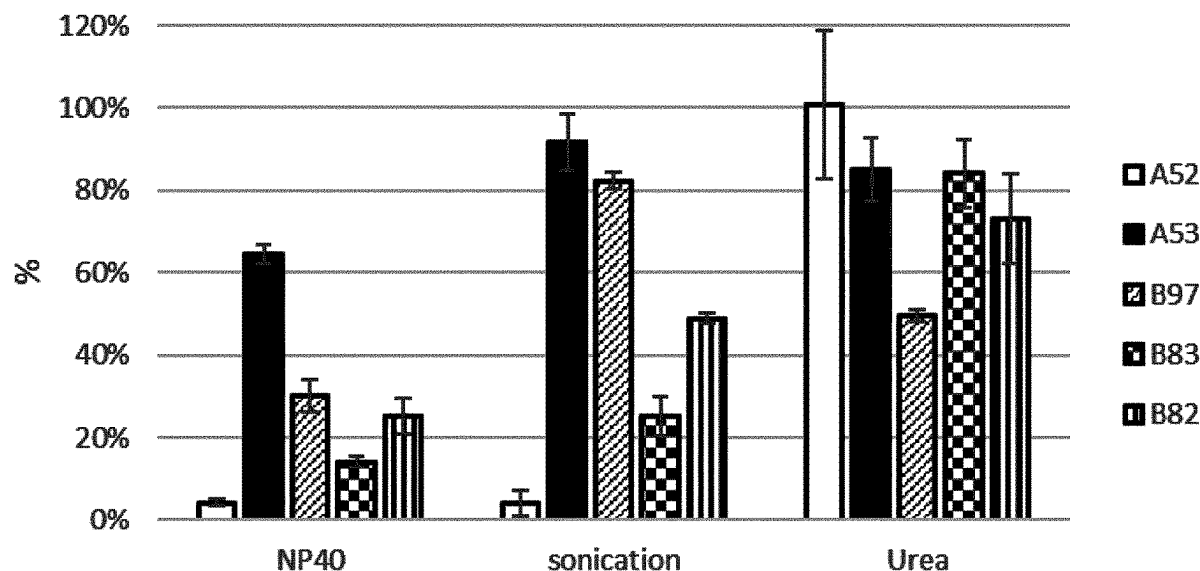


Fig. 4: Ratio of N-intein in supernatants from different extracts compared with total amount of N-intein in SDS solubilized extracts. Bars show the average solubility compared with whole cell lysate, (SDS) and the error bars show the standard deviation.

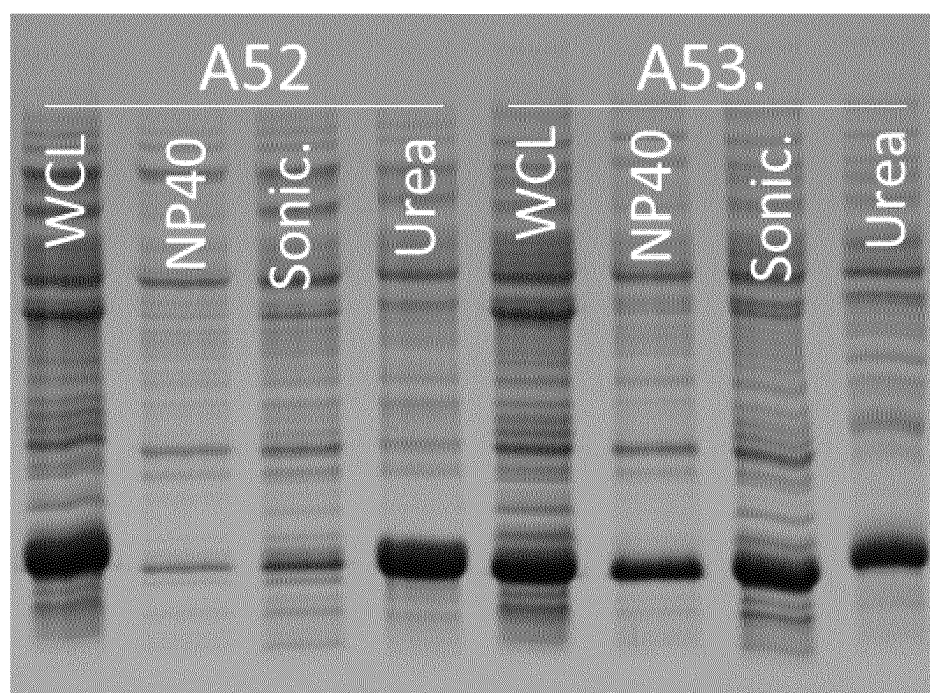


Fig. 1

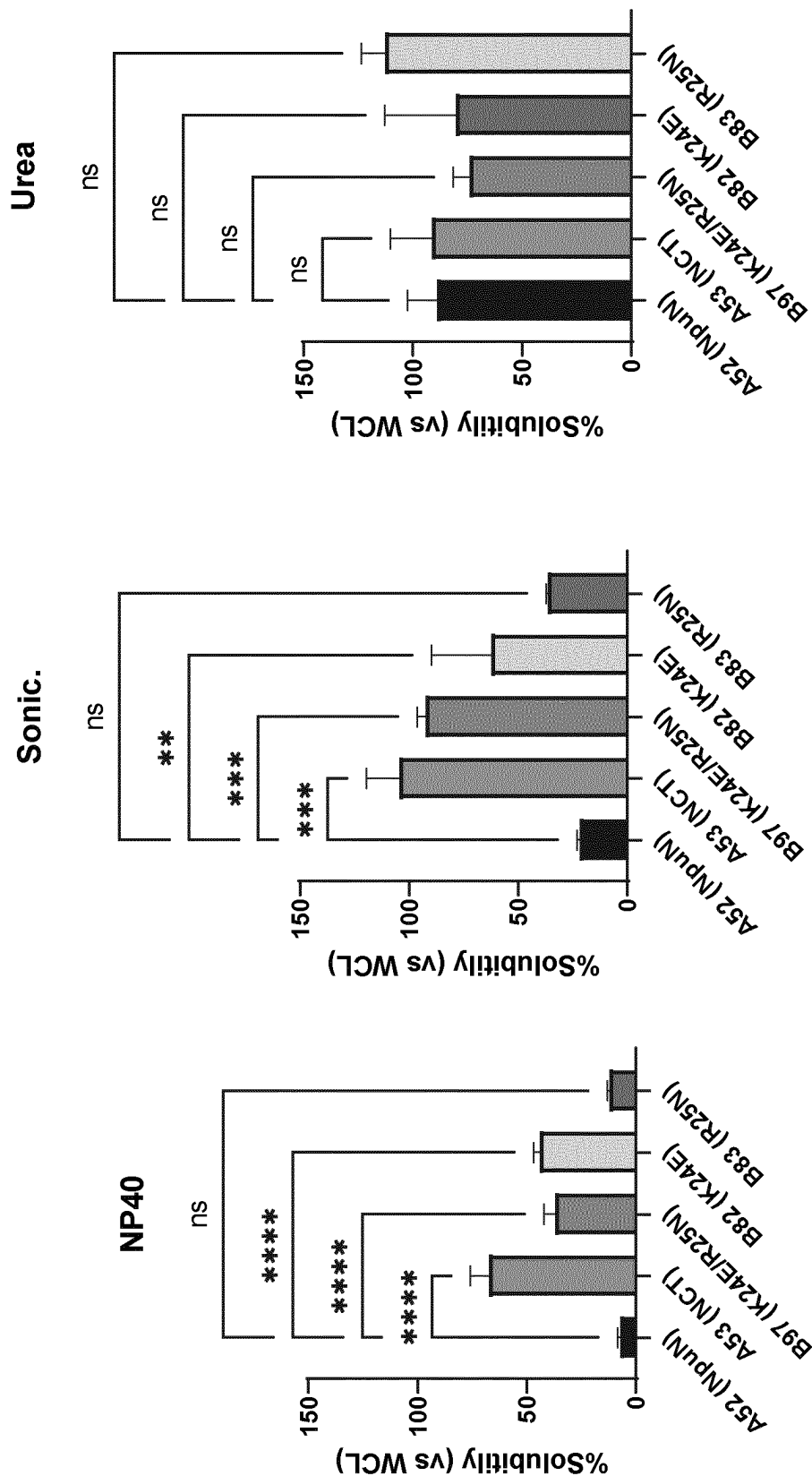


Fig. 2

Protein concentrations in supernatants after
using different extraction-methods

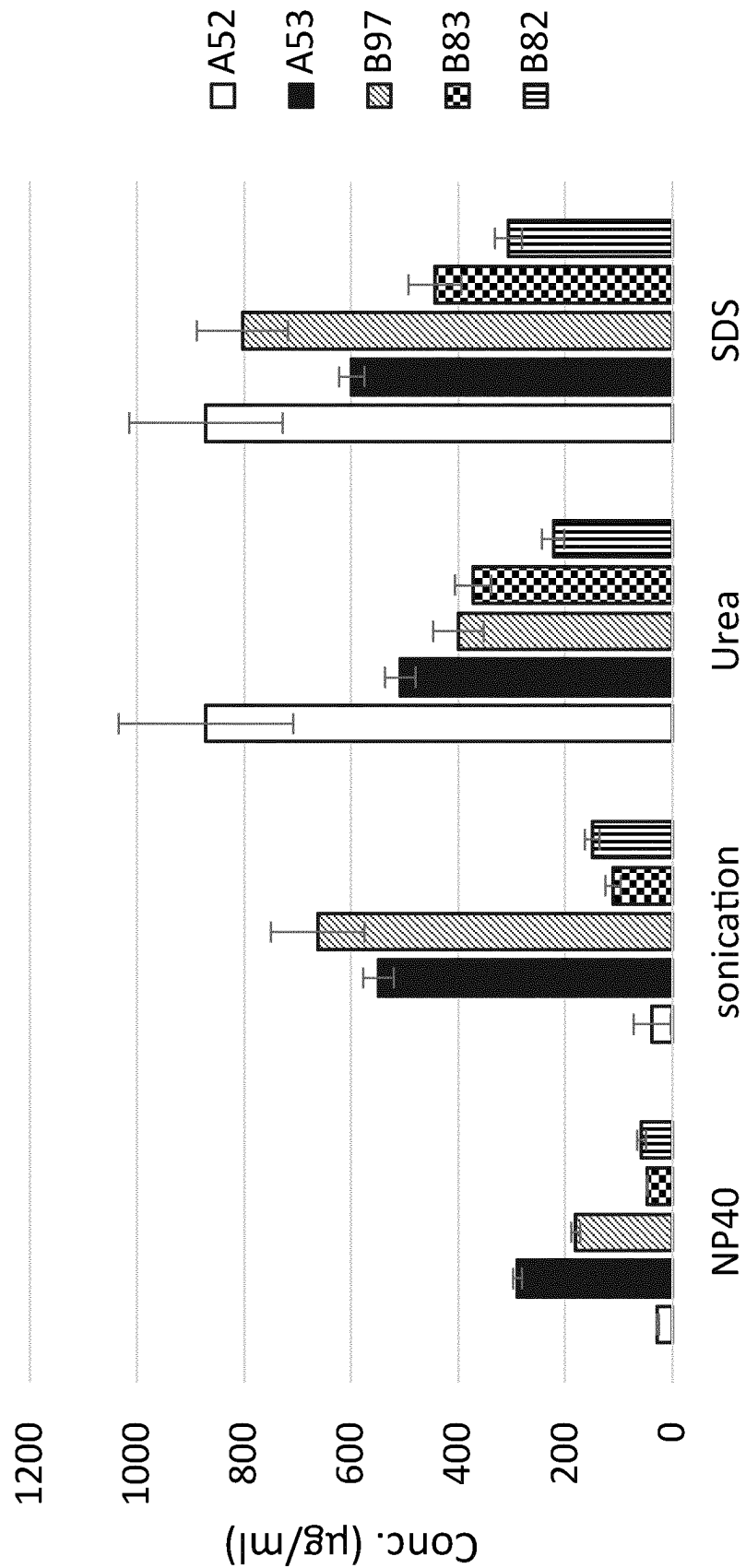


Fig. 3

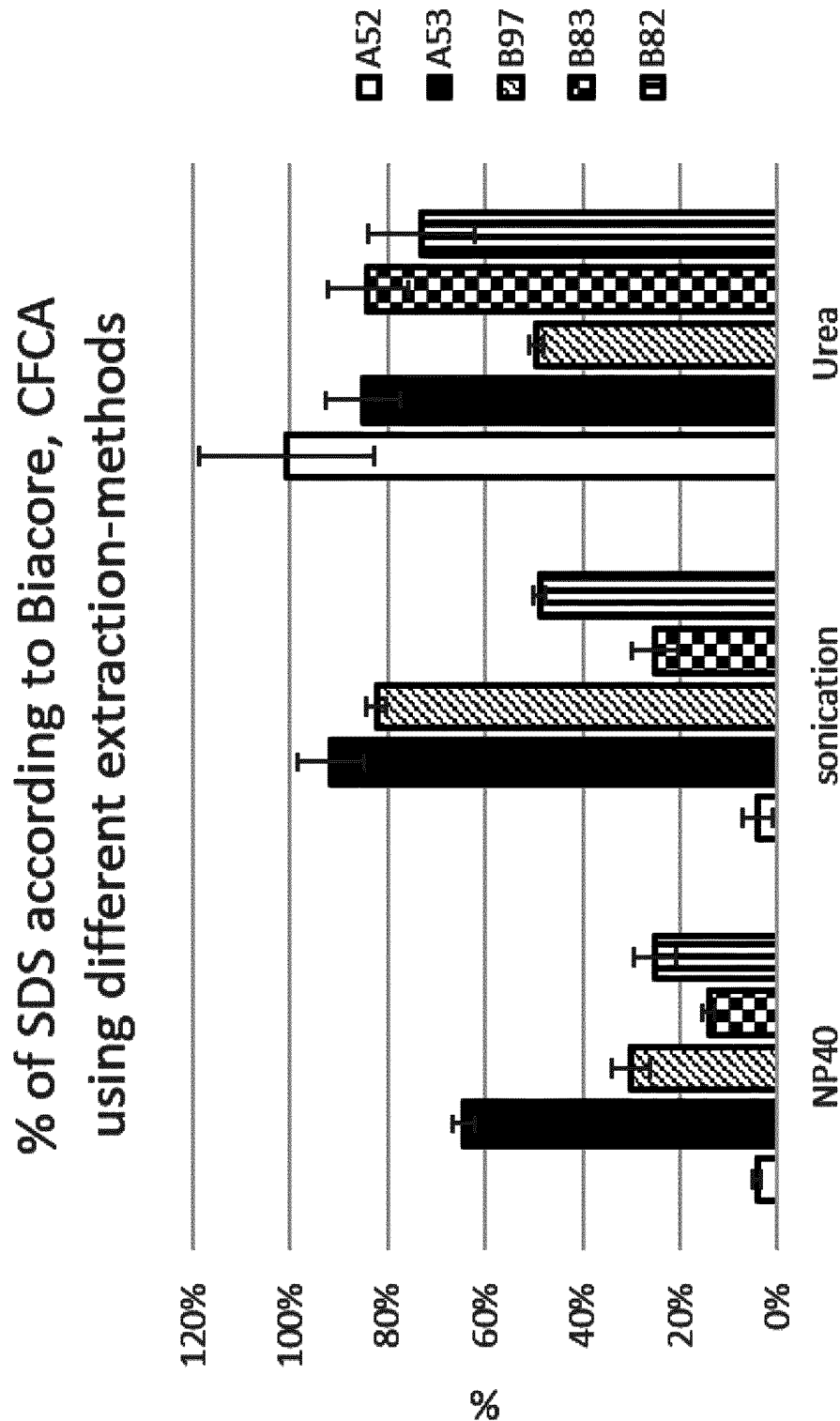


Fig. 4: Ratio of N-intein in supernatants from different extracts compared with total amount of N-intein in SDS solubilized extracts. Bars show the average solubility compared with whole cell lysate, (SDS) and the error bars show the standard deviation.

Fig. 4

IMPROVED PROTEIN PURIFICATION

FIELD OF THE INVENTION

[0001] The present invention relates to protein purification, primarily in the chromatographic field. More closely, the invention relates to affinity chromatography using a split intein system comprising a C-intein tag and N-intein ligand, wherein the N-intein ligand has high solubility and may be immobilized to a solid phase in high degree suitable for large scale protein purification.

BACKGROUND OF THE INVENTION

[0002] Inteins are protein elements expressed as in-frame insertions that interrupt enzyme sequences and catalyze their own excision and ligation of two flanking polypeptides, generating an active protein. Genetically, inteins are encoded in two distinct ways: as intact inteins, interrupting two flanking extein sequences, or as split inteins, wherein each extein and part of the intein are encoded by two different genes. While they hold great promise as bioengineering and protein purification tools, split inteins with rapid kinetic properties found in nature are dependent on specific amino acids at the intein-extein junction, severely limiting the proteins that can be fused to inteins for affinity purification and recovery of native protein sequences. In particular, the prototypical split intein DNAE from *Nostoc punctiforme* exhibits kinetic properties suitable for protein purification applications. However, its activity is dependent on phenylalanine at the +2 position in the C-extein. This dependency severely narrows and impairs its general applicability.

[0003] Inteins have been engineered to accomplish several important functions in biotechnology, including applications as self-cleaving proteins for recombinant protein purification. Split inteins are particularly promising in this regard, as they can simultaneously provide affinity ligand and self-cleavage properties. In protein purification, a target protein that is the subject of purification may be substituted for either extein. To date, the DNAE family of split inteins has shown the most promise with C-terminal cleavage protein purification approaches.

[0004] WO2014/004336 describes proteins fused to split intein N-fragments and split intein C-fragments which could be attached to a support. The solid support could be a particle, bead, resin, or a slide.

[0005] WO2014/110393 describes proteins of interest fused to a split intein C-fragment which is contacted with a split intein N-fragment and a purification tag. The N-fragment may be attached to a solid phase via the purification tag and methods for affinity purification are discussed.

[0006] U.S. Pat. No. 10,066,027 describes a protein purification system and methods of using the system. Disclosed is a split intein comprising an N-terminal intein segment, which can be immobilized, and a C-terminal intein segment, which has the property of being self-cleaving, and which can be attached to a protein of interest. The N-terminal intein segment is provided with a sensitivity enhancing motif which renders it more sensitive to extrinsic conditions.

[0007] U.S. Pat. No. 10,308,679 describes fusion proteins comprising an N-intein polypeptide and N-intein solubilization partner, and affinity matrices comprising such fusion proteins.

[0008] WO 2018/091424 describes a method for production of an affinity chromatography resin comprising an amino-terminal, (N-terminal), split intein fragment as an affinity ligand, comprising the following steps: a) expression of an N-terminal split intein fragment protein as insoluble protein in inclusion bodies in bacterial cells, preferably *E. coli*; b) harvesting said inclusion bodies; c) solubilizing said inclusion bodies and releasing expressed protein; d) binding said protein on a solid support; e) refolding said protein; f) releasing said protein from the solid support; and g) immobilizing said protein as ligands on a chromatography resin to form an affinity chromatography resin. This procedure enables immobilization a ligand density of 2-10 mg/ml resin.

[0009] As described above, split inteins have been used for protein purification using a combined affinity tag and tag cleavage mechanism. However, the utility of such systems, is limited by several factors. First, there is the amino acid requirements at the splice junction of the intended product, i.e. the requirement of Phe in the +2 position of the C-extein, to effect cleavage and attain purification of tag-less proteins. Recombinant protein production without extraneous amino acid on the N-terminus is highly desirable. Second, the protein releasing cleavage has to be sufficiently fast and provide an acceptable yield. Third, there is a solubility requirement of the split intein N- or C-fragment for attachment thereof to a solid support. Fourth, hitherto there are no available split intein systems suitable for large scale purification of tag-less proteins.

[0010] One way to improve the solubility is by attaching a solubility fusion-tag to the split-inteins, (U.S. Pat. No. 10,308,679). The development of methods for protein expression and purification is commonly facilitated by the use of fusion tags that offer the possibility to standardize protocols for purification, simplify the detection and increase the solubility of a target protein. Fusion tags can however interfere with protein function and structure. It is therefore advantageous to remove fusion tags prior to usage. A large fusion tag relative to a target protein also results in an increased metabolic burden for the host cells expressing these fusion proteins, since additional energy is spent on the fusion tag.

[0011] A different approach to increase the efficiency of producing highly insoluble split-inteins is by solubilizing proteins with denaturing chemical reagents followed by a refolding process, (U.S. application Ser. No. 16/348,534) to regain bioactive protein. Attempts have been made to understand the technical aspects of various methods used for protein refolding along with their advantages and limitations, but usually the efficiency and yield in such methods is very difficult to predict and has to be determined by empirical studies for each protein. A common problem in refolding methods is the formation of protein aggregates when the denaturing chemicals are being removed or diluted during refolding. These aggregates lower the yield in the process and adds complexity during the subsequent purification steps in a production process. Moreover, the denaturing chemicals are usually a burden to the environment and needs to be properly handled.

SUMMARY OF THE INVENTION

[0012] The present invention overcomes the disadvantages within prior art and provides a N-intein polypeptide, which is soluble without the need for a solubility fusion-tag

and that can be produced in an industrial scale in an environmentally friendly production process for subsequent use in affinity purification processes.

[0013] In particular, the invention provides a method to increase the solubility of the prototypical split intein DnaE from *Nostoc punctiforme*, which exhibits kinetic properties suitable for protein purification applications. However, the method is not limited to DnaE split intein from *Nostoc punctiforme* but is also applicable for homologous split inteins from other species. Solubility refers to a protein that after a substitution of one or preferably two amino acids in the polypeptide chain has a higher ratio of soluble N-intein expressed in *E. coli* relative to the soluble N-intein ratio in the absence of these amino acid substitutions.

[0014] The present invention provides N-intein protein variants of native split inteins or consensus sequences derived from inteins/split inteins wherein the N-intein protein variant has one or more mutations for increased solubility.

[0015] In a first aspect, the invention relates to an N-intein variant derived from native *Nostoc punctiforme* (Npu) or sequences having at least 95% homology therewith comprising at least one amino acid substitution of a native split intein wherein the N-intein protein variant sequence includes a mutation in at least position 24 and/or position 25 as measured from the initial catalytic cysteine and wherein the substituted amino acid provides increased solubility in aqueous buffers compared to the native N-intein protein sequence or a consensus N-intein sequence. The invention also encompasses inteins which have a naturally occurring E in position 24, such as N-inteins from *Limnorafis robusta*.

[0016] Preferably the substituted amino acid(s) that provide increased solubility is a non-positive amino acid. In a preferred embodiment the substituted amino acid that provide increased solubility is K24E. In another embodiment the substituted amino acid that provide increased solubility is R25N. In a most preferred embodiment the N-intein comprises both these mutations.

[0017] The invention relates to an N-intein protein variant of the wildtype N-intein domain of *Nostoc punctiforme* (Npu) wherein the wildtype Npu N-intein domain comprises the following sequence: CLSYETEILTVEYGLLPIG-KIVEKRIECTVYSVDNNG-NIYTQPVQAQWHDRGEQEVFEY CLEDGSLIRATKDHKFMFTVDGQMLPIDEIFERELDLMRV (SEQ ID NO 1) (construct A52 in the Examples), wherein the protein variant comprises an amino acid substitution from K to E in position 24 and R to N in position 25 (construct B97 in the Examples) to increase solubility in aqueous buffers to minimize formation of inclusion bodies, wherein optionally one or more C (Cys) are mutated to non-Cysteine residues, preferably S (Ser) or A (Ala). Further constructs encompassed by the invention are described in the example section below.

[0018] The N-intein protein variant as described above has solubility in aqueous buffer of at least 10-40% soluble N-intein with a single-point mutation of R at position 25, preferred N or non-positive amino acid; at least 46-52% soluble N-intein with a single-point mutation of K at position 24, preferred E or non-positive amino acid; and at least 76-88% soluble N-intein with mutations at positions 24 and 25, preferred K24E and R25N or non-positive amino acids.

[0019] The N-intein variant may be coupled to solid phase, such as a membrane, fiber, particle, bead or chip, such as chromatography resin of natural or synthetic origin.

[0020] The solid phase may optionally be provided with embedded magnetic particles. In an alternative embodiment the solid phase is a non-diffusion limited resin/fibrous material. According to the invention 0.2-2 μ mole/ml N-intein is coupled per ml solid phase, preferably chromatography resin (ml swollen gel).

[0021] In a second aspect the invention relates to a split intein system comprising a N-intein as described above and a C-intein sequence which is co-expressed with a POI (protein of interest). The C-intein acts as a tag on the POI for binding to the N-intein attached to solid phase. After binding, the POI is cleaved off from the combined N-intein and C-intein and delivering a tagless POI. The C-intein variant is a split intein C-intein sequence or engineered variants thereof. A preferred C-intein sequence is mentioned in WO2021/099607 A1.

[0022] The POI's may be any recombinant proteins: proteins requiring native or near native N-terminal sequences, for example therapeutic protein candidates, biologics, antibody fragments, antibody mimetics, enzymes, recombinant proteins or peptides, such as growth factors, cytokines, chemokines, hormones, antigen (viral, bacterial, yeast, mammalian) production, vaccine production, cell surface receptors, fusion proteins.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 shows a SDS-PAGE analysis of supernatants from different constructs after extraction using different techniques.

[0024] FIG. 2 shows solubility of different constructs determined after densitometric evaluation of SDS-PAGE analysis. Extracts from three different cell-cultures for each construct were analysed. Bars show the average solubility compared with whole cell lysate, (SDS) and the error bars show the standard deviation.

[0025] FIG. 3 shows N-intein concentrations in supernatants from different extracts determined by Biacore CFCA analysis. Extracts from three different cell-cultures for each construct were analysed. Bars show the average concentration and the error bars show the standard deviation.

[0026] FIG. 4 shows the ratio of N-intein in supernatants from extracts of different constructs using different extraction methods, compared as % to total amount of N-intein solubilized by SDS and heating.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

[0027] As used in the specification and the appended claims, the singular forms "a," "an" and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a functional group," "an alkyl," or "a residue" includes mixtures of two or more such functional groups, alkyls, or residues, and the like.

[0028] Ranges can be expressed herein as from "about" one particular value, and/or to "about" another particular value. When such a range is expressed, a further aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as

approximations, by use of the antecedent “about,” it will be understood that the particular value forms a further aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint. It is also understood that there are a number of values disclosed herein, and that each value is also herein disclosed as “about” that particular value in addition to the value itself.

[0029] A weight percent (wt. %) of a component, unless specifically stated to the contrary, is based on the total weight of the formulation or composition in which the component is included.

[0030] As used herein, the terms “optional” or “optionally” means that the subsequently described event or circumstance can or can not occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

[0031] The term “contacting” as used herein refers to bringing two biological entities together in such a manner that the compound can affect the activity of the target, either directly; i.e., by interacting with the target itself, or indirectly; i.e., by interacting with another molecule, co-factor, factor, or protein on which the activity of the target is dependent. “Contacting” can also mean facilitating the interaction of two biological entities, such as peptides, to bond covalently or otherwise.

[0032] The term “peptide”, “polypeptides” and “protein” are used interchangeably herein and include proteins and fragments thereof. Polypeptides are disclosed herein as amino acid residue sequences. Those sequences are written left to right in the direction from the amino to the carboxy terminus. In accordance with standard nomenclature, amino acid residue sequences are denominated by either a three letter or a single letter code as indicated as follows: Alanine (Ala, A), Arginine (Arg, R), Asparagine (Asn, N), Aspartic Acid (Asp, D), Cysteine (Cys, C), Glutamine (Gln, Q), Glutamic Acid (Glu, E), Glycine (Gly, G), Histidine (His, H), Isoleucine (Ile, I), Leucine (Leu, L), Lysine (Lys, K), Methionine (Met, M), Phenylalanine (Phe, F), Proline (Pro, P), Serine (Ser, S), Threonine (Thr, T), Tryptophan (Trp, W), Tyrosine (Tyr, Y), and Valine (Val, V). Peptides include any oligopeptide, polypeptide, gene product, expression product, or protein. A peptide is comprised of consecutive amino acids and encompasses naturally occurring or synthetic molecules.

[0033] In addition, as used herein, the term “peptide” refers to amino acids joined to each other by peptide bonds or modified peptide bonds, e.g., peptide isosteres, etc. and may contain modified amino acids other than the 20 gene-encoded amino acids. The peptides can be modified by either natural processes, such as post-translational processing, or by chemical modification techniques which are well known in the art. Modifications can occur anywhere in the peptide, including the peptide backbone, the amino acid side-chains and the amino or carboxyl termini. The same type of modification can be present in the same or varying degrees at several sites in a given polypeptide. Also, a given peptide can have many types of modifications. Modifications include, without limitation, linkage of distinct domains or motifs, acetylation, acylation, ADP-ribosylation, amidation, covalent cross-linking or cyclization, covalent attachment of flavin, covalent attachment of a heme moiety, covalent attachment of a nucleotide or nucleotide derivative, covalent attachment of a lipid or lipid derivative, covalent attachment

of a phosphatidylinositol, disulfide bond formation, demethylation, formation of cysteine or pyroglutamate, formylation, gamma-carboxylation, glycosylation, GPI anchor formation, hydroxylation, iodination, methylation, myristoylation, oxidation, pergylation, proteolytic processing, phosphorylation, prenylation, racemization, selenoylation, sulfation, and transfer-RNA mediated addition of amino acids to protein such as arginylation. (See *Proteins—Structure and Molecular Properties* 2nd Ed., T. E. Creighton, W. H. Freeman and Company, New York (1993); *Posttranslational Covalent Modification of Proteins*, B. C. Johnson, Ed., Academic Press, New York, pp. 1-12 (1983)).

[0034] As used herein, “variant” refers to a molecule that retains a biological activity that is the same or substantially similar to that of the original sequence. The variant may be from the same or different species or be a synthetic sequence based on a natural or prior molecule. Moreover, as used herein, “variant” refers to a molecule having a structure attained from the structure of a parent molecule (e.g., a protein or peptide disclosed herein) and whose structure or sequence is sufficiently similar to those disclosed herein that based upon that similarity, would be expected by one skilled in the art to exhibit the same or similar activities and utilities compared to the parent molecule. For example, substituting specific amino acids in a given peptide can yield a variant peptide with similar activity to the parent.

[0035] In the context of the present invention, a substitution in a variant protein is indicated as: [original amino acid/position in sequence/substituted amino acid].

[0036] As used herein, the term “protein of interest (POI)” includes any synthetic or naturally occurring protein or peptide. The term therefore encompasses those compounds traditionally regarded as drugs, vaccines, and biopharmaceuticals including molecules such as proteins, peptides, and the like. Examples of therapeutic agents are described in well-known literature references such as the Merck Index (14th edition), the Physicians’ Desk Reference (64th edition), and The Pharmacological Basis of Therapeutics (1st edition), and they include, without limitation, medicaments; substances used for the treatment, prevention, diagnosis, cure or mitigation of a disease or illness; substances that affect the structure or function of the body, or pro-drugs, which become biologically active or more active after they have been placed in a physiological environment.

[0037] As used herein, “isolated peptide” or “purified peptide” is meant to mean a peptide (or a fragment thereof) that is substantially free from the materials with which the peptide is normally associated in nature, or from the materials with which the peptide is associated in an artificial expression or production system, including but not limited to an expression host cell lysate, growth medium components, buffer components, cell culture supernatant, or components of a synthetic in vitro translation system. The peptides disclosed herein, or fragments thereof, can be obtained, for example, by extraction from a natural source (for example, a mammalian cell), by expression of a recombinant nucleic acid encoding the peptide (for example, in a cell or in a cell-free translation system), or by chemically synthesizing the peptide. In addition, peptide fragments may be obtained by any of these methods, or by cleaving full length proteins and/or peptides.

[0038] The word “or” as used herein means any one member of a particular list and also includes any combination of members of that list.

[0039] The phrase “nucleic acid” as used herein refers to a naturally occurring or synthetic oligonucleotide or polynucleotide, whether DNA or RNA or DNA-RNA hybrid, single-stranded or double-stranded, sense or antisense, which is capable of hybridization to a complementary nucleic acid by Watson-Crick base-pairing. Nucleic acids of the invention can also include nucleotide analogs (e.g., BrdU), and non-phosphodiester internucleoside linkages (e.g., peptide nucleic acid (PNA) or thiodiester linkages). In particular, nucleic acids can include, without limitation, DNA, RNA, cDNA, gDNA, ssDNA, dsDNA or any combination thereof.

[0040] As used herein, “extein” refers to the portion of an intein-modified protein that is not part of the intein and which can be spliced or cleaved upon excision of the intein.

[0041] “Intein” refers to an in-frame intervening sequence in a protein. An intein can catalyze its own excision from the protein through a post-translational protein splicing process to yield the free intein and a mature protein. An intein can also catalyze the cleavage of the intein-extein bond at either the intein N-terminus, or the intein C-terminus, or both of the intein-extein termini. As used herein, “intein” encompasses mini-inteins, modified or mutated inteins, and split inteins.

[0042] As used herein, the term “split intein” refers to any intein in which one or more peptide bond breaks exists between the N-terminal intein segment and the C-terminal intein segment such that the N-terminal and C-terminal intein segments become separate molecules that can non-covalently reassociate, or reconstitute, into an intein that is functional for splicing or cleaving reactions. Any catalytically active intein, or fragment thereof, may be used to derive a split intein for use in the systems and methods disclosed herein. For example, in one aspect the split intein may be derived from a eukaryotic intein. In another aspect, the split intein may be derived from a bacterial intein. In another aspect, the split intein may be derived from an archaeal intein. Preferably, the split intein so-derived will possess only the amino acid sequences essential for catalyzing splicing reactions.

[0043] As used herein, the “N-terminal intein segment” or “N-intein” refers to any intein sequence that comprises an N-terminal amino acid sequence that is functional for splicing and/or cleaving reactions when combined with a corresponding C-terminal intein segment. An N-terminal intein segment thus also comprises a sequence that is spliced out when splicing occurs. An N-terminal intein segment can comprise a sequence that is a modification of the N-terminal portion of a naturally occurring (native) intein sequence. Non-intein residues can also be genetically fused to intein segments to provide additional functionality, such as the ability to be affinity purified or to be covalently immobilized.

[0044] As used herein, the “C-terminal intein segment” or “C-intein” refers to any intein sequence that comprises a C-terminal amino acid sequence that is functional for splicing or cleaving reactions when combined with a corresponding N-terminal intein segment. In one aspect, the C-terminal intein segment comprises a sequence that is spliced out when splicing occurs. In another aspect, the C-terminal intein segment is cleaved from a peptide sequence fused to its C-terminus. The sequence which is cleaved from the C-terminal intein’s C-terminus is referred to herein as a “protein of interest POI” is discussed in more detail below.

A C-terminal intein segment can comprise a sequence that is a modification of the C-terminal portion of a naturally occurring (native) intein sequence. For example, a C terminal intein segment can comprise additional amino acid residues and/or mutated residues so long as the inclusion of such additional and/or mutated residues does not render the C-terminal intein segment non-functional for splicing or cleaving.

[0045] A consensus sequence is a sequence of DNA, RNA, or protein that represents aligned, related sequences. The consensus sequence of the related sequences can be defined in different ways, but is normally defined by the most common nucleotide(s) or amino acid residue(s) at each position.

[0046] As used herein, the term “splice” or “splices” means to excise a central portion of a polypeptide to form two or more smaller polypeptide molecules. In some cases, splicing also includes the step of fusing together two or more of the smaller polypeptides to form a new polypeptide. Splicing can also refer to the joining of two polypeptides encoded on two separate gene products through the action of a split intein.

[0047] As used herein, the term “cleave” or “cleaves” means to divide a single polypeptide to form two or more smaller polypeptide molecules. In some cases, cleavage is mediated by the addition of an extrinsic endopeptidase, which is often referred to as “proteolytic cleavage”. In other cases, cleaving can be mediated by the intrinsic activity of one or both of the cleaved peptide sequences, which is often referred to as “self-cleavage”. Cleavage can also refer to the self-cleavage of two polypeptides that is induced by the addition of a non-proteolytic third peptide, as in the action of split intein system described herein.

[0048] By the term “fused” is meant covalently bonded to. For example, a first peptide is fused to a second peptide when the two peptides are covalently bonded to each other (e.g., via a peptide bond).

[0049] As used herein an “isolated” or “substantially pure” substance is one that has been separated from components which naturally accompany it. Typically, a polypeptide is substantially pure when it is at least 50% (e.g., 60%, 70%, 80%, 90%, 95%, and 99%) by weight free from the other proteins and naturally-occurring organic molecules with which it is naturally associated.

[0050] Herein, “bind” or “binds” means that one molecule recognizes and adheres to another molecule in a sample, but does not substantially recognize or adhere to other molecules in the sample. One molecule “specifically binds” another molecule if it has a binding affinity greater than about 10^5 to 10^6 liters/mole for the other molecule.

[0051] Nucleic acids, nucleotide sequences, proteins or amino acid sequences referred to herein can be isolated, purified, synthesized chemically, or produced through recombinant DNA technology. All of these methods are well known in the art.

[0052] As used herein, the terms “modified” or “mutated,” as in “modified intein” or “mutated intein,” refer to one or more modifications in either the nucleic acid or amino acid sequence being referred to, such as an intein, when compared to the native, or naturally occurring structure. Such modification can be a substitution, addition, or deletion. The modification can occur in one or more amino acid residues or one or more nucleotides of the structure being referred to, such as an intein.

[0053] As used herein, the term “modified peptide”, “modified protein” or “modified protein of interest” or “modified target protein” refers to a protein which has been modified.

[0054] As used herein, “operably linked” refers to the association of two or more biomolecules in a configuration relative to one another such that the normal function of the biomolecules can be performed. In relation to nucleotide sequences, “operably linked” refers to the association of two or more nucleic acid sequences, by means of enzymatic ligation or otherwise, in a configuration relative to one another such that the normal function of the sequences can be performed. For example, the nucleotide sequence encoding a pre-sequence or secretory leader is operably linked to a nucleotide sequence for a polypeptide if it is expressed as a pre-protein that participates in the secretion of the polypeptide; a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the coding sequence; and a ribosome binding site is operably linked to a coding sequence if it is positioned so as to facilitate translation of the sequence.

[0055] “Sequence homology” can refer to the situation where nucleic acid or protein sequences are similar because they have a common evolutionary origin. “Sequence homology” can indicate that sequences are very similar. Sequence similarity is observable; homology can be based on the observation. “Very similar” can mean at least 70% identity, homology or similarity; at least 75% identity, homology or similarity; at least 80% identity, homology or similarity; at least 85% identity, homology or similarity; at least 90% identity, homology or similarity; such as at least 93% or at least 95% or even at least 97% identity, homology or similarity. The nucleotide sequence similarity or homology or identity can be determined using the “Align” program of Myers et al. (1988) CABIOS 4:11-17 and available at NCBI. Additionally or alternatively, amino acid sequence similarity or identity or homology can be determined using the BlastP program (Altschul et al. Nucl. Acids Res. 25:3389-3402), and available at NCBI. Alternatively or additionally, the terms “similarity” or “identity” or “homology,” for instance, with respect to a nucleotide sequence, are intended to indicate a quantitative measure of homology between two sequences.

[0056] Alternatively or additionally, “similarity” with respect to sequences refers to the number of positions with identical nucleotides divided by the number of nucleotides in the shorter of the two sequences wherein alignment of the two sequences can be determined in accordance with the Wilbur and Lipman algorithm. (1983) Proc. Natl. Acad. Sci. USA 80:726. For example, using a window size of 20 nucleotides, a word length of 4 nucleotides, and a gap penalty of 4, and computer-assisted analysis and interpretation of the sequence data including alignment can be conveniently performed using commercially available programs (e.g., Intelligenetics™ Suite, Intelligenetics™ Inc. CA). When RNA sequences are said to be similar, or have a degree of sequence identity with DNA sequences, thymidine (T) in the DNA sequence is considered equal to uracil (U) in the RNA sequence. The following references also provide algorithms for comparing the relative identity or homology or similarity of amino acid residues of two proteins, and additionally or alternatively with respect to the foregoing, the references can be used for determining percent homology or identity or similarity. Needleman et al. (1970) J. Mol.

Biol. 48:444-453; Smith et al. (1983) Advances App. Math. 2:482-489; Smith et al. (1981) Nuc. Acids Res. 11:2205-2220; Feng et al. (1987) J. Molec. Evol. 25:351-360; Higgins et al. (1989) CABIOS 5:151-153; Thompson et al. (1994) Nuc. Acids Res. 22:4673-480; and Devereux et al. (1984) 12:387-395. “Stringent hybridization conditions” is a term which is well known in the art; see, for example, Sambrook, “Molecular Cloning, A Laboratory Manual” second ed., CSH Press, Cold Spring Harbor, 1989; “Nucleic Acid Hybridization, A Practical Approach”, Hames and Higgins eds., IRL Press, Oxford, 1985; see also FIG. 2 and description thereof herein wherein there is a sequence comparison.

[0057] The term “buffer” or “buffered solution” refers to solutions which resist changes in pH by the action of its conjugate acid-base range.

[0058] The term “loading buffer” or “equilibrium buffer” refers to the buffer containing the salt or salts which is mixed with the protein preparation for loading the protein preparation onto a column. This buffer is also used to equilibrate the column before loading, and to wash to column after loading the protein.

[0059] The term “wash buffer” is used herein to refer to the buffer that is passed over a column (for example) following loading of a protein of interest (such as one coupled to a C-terminal intein fragment, for example) and prior to elution of the protein of interest. The wash buffer may serve to remove one or more contaminants without substantial elution of the desired protein.

[0060] The term “elution buffer” refers to the buffer used to elute the desired protein from the column. As used herein, the term “solution” refers to either a buffered or a non-buffered solution, including water.

[0061] The term “washing” means passing an appropriate buffer through or over a solid support, such as a chromatographic resin.

[0062] The term “eluting” a molecule (e.g. a desired protein or contaminant) from a solid support means removing the molecule from such material.

[0063] The term “contaminant” or “impurity” refers to any foreign or objectionable molecule, particularly a biological macromolecule such as a DNA, an RNA, or a protein, other than the protein being purified, that is present in a sample of a protein being purified. Contaminants include, for example, other proteins from cells that express and/or secrete the protein being purified.

[0064] The term “separate” or “isolate” as used in connection with protein purification refers to the separation of a desired protein from a second protein or other contaminant or mixture of impurities in a mixture comprising both the desired protein and a second protein or other contaminant or impurity mixture, such that at least the majority of the molecules of the desired protein are removed from that portion of the mixture that comprises at least the majority of the molecules of the second protein or other contaminant or mixture of impurities.

[0065] The term “purify” or “purifying” a desired protein from a composition or solution comprising the desired protein and one or more contaminants means increasing the degree of purity of the desired protein in the composition or solution by removing (completely or partially) at least one contaminant from the composition or solution.

N-Intein Protein Variants

[0066] The invention relates to affinity chromatography and affinity tag cleavage mechanisms in a single step using a split intein system according to the invention which cleaves with broad amino acid tolerance to generate a tag less protein of interest (POI) as end product. The two halves of the intein are the affinity ligand (N-intein) and the affinity tag (C-intein) and they associate rapidly. Immobilizing one half (N-intein) on a chromatography resin enables the capture of the other half (C-intein) coupled to the POI from solution. In the presence of Zn^{2+} ions, the cleavage reaction is inhibited, enabling a stable complex to form while impurities are washed away. After impurities are eliminated, a chelator or reducing agent is added, and the cleavage reaction proceeds, enabling collection of the POI, while the intein tag remains bound non-covalently to the cognate intein linked to the chromatography resin.

[0067] Preferably the invention provides N-intein protein variant sequences of native split inteins or consensus sequences derived from native inteins and split inteins wherein, the N-intein variant is modified as compared to the native sequence or consensus sequence to provide increased solubility by having mutations in position 24 and/or 25. These positions are calculated according to conventional clustal alignment with native split inteins starting from the initial catalytical cysteine which is number 1.

[0068] Native intein are known in the art. A list of inteins is found in Table 1 below. All inteins have the potential to be made into split inteins while some inteins naturally exist in split form. All of the inteins found in the table either exist as split inteins or have the potential to be made into split inteins modified in accordance with the invention at position 24 and/or 25 such that the increased solubility is achieved compared to the native sequences.

TABLE 1

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
<u>Eucarya</u>		
APMV Pol	<i>Acanthamoeba polyphaga</i> Mimivirus	isolate = "Rowbotham-Bradford", Virus, infects Amoeboae, taxon: 212035
Abr PRP8	<i>Aspergillus brevipes</i> FRR2439	Fungi, ATCC 16899, taxon: 75551
Aca-G186AR PRP8	<i>Ajellomyces capsulatus</i> G186AR	Taxon: 447093, strain G186AR
Aca-H143 PRP8	<i>Ajellomyces capsulatus</i> H143	Taxon: 544712
Aca-JER2004 PRP8	<i>Ajellomyces capsulatus</i> (anamorph: <i>Histoplasma capsulatum</i>)	strain = JER2004, taxon: 5037, Fungi
Aca-NAm1 PRP8	<i>Ajellomyces capsulatus</i> NAm1	strain = "NAm1", taxon: 339724
Ade-ER3 PRP8	<i>Ajellomyces dermatitidis</i> ER-3	Human fungal pathogen, taxon: 559297
Ade-SLH14081 PRP8	<i>Ajellomyces dermatitidis</i> SLH14081,	Human fungal pathogen
Afu-Af293 PRP8	<i>Aspergillus fumigatus</i> var. <i>ellipticus</i> , strain Af293	Human pathogenic fungus, taxon: 330879
Afu-FRR0163 PRP8	<i>Aspergillus fumigatus</i> strain FRR0163	Human pathogenic fungus, taxon: 5085
Afu-NRRL5109 PRP8	<i>Aspergillus fumigatus</i> var. <i>ellipticus</i> , strain NRRL 5109	Human pathogenic fungus, taxon: 41121
Agi-NRRL6136 PRP8	<i>Aspergillus giganteus</i> Strain NRRL 6136	Fungus, taxon: 5060
Ani-FGSCA4 PRP8	<i>Aspergillus nidulans</i> FGSC A	Filamentous fungus, taxon: 227321
Avi PRP8	<i>Aspergillus viridinutans</i> strain FRR0577	Fungi, ATCC 16902, taxon: 75553
Bci PRP8	<i>Botrytis cinerea</i> (teleomorph of <i>Botryotinia fuckeliana</i> B05.10)	Plant fungal pathogen
Bde-JEL197 RPB2	<i>Batrachochytrium dendrobatidis</i> JEL197	Chytrid fungus, isolate = "AFTOL-ID 21", taxon: 109871
Bde-JEL423 PRP8-1	<i>Batrachochytrium dendrobatidis</i> JEL423	Chytrid fungus, isolate JEL423, taxon 403673
Bde-JEL423 PRP8-2	<i>Batrachochytrium dendrobatidis</i> JEL423	Chytrid fungus, isolate JEL423, taxon 403673
Bde-JEL423 RPC2	<i>Batrachochytrium dendrobatidis</i> JEL423	Chytrid fungus, isolate JEL423, taxon 403673
Bde-JEL423 eIF-5B	<i>Batrachochytrium dendrobatidis</i> JEL423	Chytrid fungus, isolate JEL423, taxon 403673
Bfu-B05 PRP8	<i>Botryotinia fuckeliana</i> B05.10	Taxon: 332648
CIV RIR1	Chilo iridescent virus	dsDNA eucaryotic virus, taxon: 10488
CV-NY2A ORF212392	<i>Chlorella</i> virus NY2A infects <i>Chlorella</i> NC64A, which infects <i>Paramecium bursaria</i>	dsDNA eucaryotic virus, taxon: 46021, Family Phycodnaviridae
CV-NY2A RIR1	<i>Chlorella</i> virus NY2A infects <i>Chlorella</i> NC64A, which infects <i>Paramecium bursaria</i>	dsDNA eucaryotic virus, taxon: 46021, Family Phycodnaviridae

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
CZIV RIR1	<i>Costelytra zealandica</i> iridescent virus	dsDNA eucaryotic virus, Taxon: 68348
Cba-WM02.98 PRP8	<i>Cryptococcus bacillisporus</i> strain WM02.98 (aka <i>Cryptococcus neoformans gattii</i>)	Yeast, human pathogen, taxon: 37769
Cba-WM728 PRP8	<i>Cryptococcus bacillisporus</i> strain WM728	Yeast, human pathogen, taxon: 37769
Ceu ClpP	<i>Chlamydomonas eugametos</i> (chloroplast)	Green alga, taxon: 3053
Cga PRP8	<i>Cryptococcus gattii</i> (aka <i>Cryptococcus bacillisporus</i>)	Yeast, human pathogen
Cgl VMA	<i>Candida glabrata</i>	Yeast, taxon: 5478
Cla PRP8	<i>Cryptococcus laurentii</i> strain CBS139	Fungi, Basidiomycete yeast, taxon: 5418
Cmo ClpP	<i>Chlamydomonas moewusii</i> , strain UTEX 97	Green alga, chloroplast gene, taxon: 3054
Cmo RPB2 (RpoBb)	<i>Chlamydomonas moewusii</i> , strain UTEX 97	Green alga, chloroplast gene, taxon: 3054
Cne-A PRP8 (Fne-A PRP8)	<i>Filobasidiella neoformans</i> (<i>Cryptococcus neoformans</i>) Serotype A, PHLS_8104	Yeast, human pathogen
Cne-AD PRP8 (Fne- AD PRP8)	<i>Cryptococcus neoformans</i> (<i>Filobasidiella neoformans</i>), Serotype AD, CBS132).	Yeast, human pathogen, ATCC32045, taxon: 5207
Cne-JEC21 PRP8	<i>Cryptococcus neoformans</i> var. <i>neoformans</i> JEC21	Yeast, human pathogen, serotype = "D" taxon: 214684
Cpa ThrRS	<i>Candida parapsilosis</i> , strain CLIB214	Yeast, Fungus, taxon: 5480
Cre RPB2	<i>Chlamydomonas reinhardtii</i> (nucleus)	Green algae, taxon: 3055
CroV Pol	<i>Cafeteria roenbergensis</i> virus BV- PW1	taxon: 693272, Giant virus infecting marine heterotrophic nanoflagellate
CroV RIR1	<i>Cafeteria roenbergensis</i> virus BV- PW1	taxon: 693272, Giant virus infecting marine heterotrophic nanoflagellate
CroV RPB2	<i>Cafeteria roenbergensis</i> virus BV- PW1	taxon: 693272, Giant virus infecting marine heterotrophic nanoflagellate
CroV Top2	<i>Cafeteria roenbergensis</i> virus BV- PW1	taxon: 693272, Giant virus infecting marine heterotrophic nanoflagellate
Cst RPB2	<i>Coelomomyces stegomyiae</i>	Chytrid fungus, isolate = "AFTOL-ID 18", taxon: 143960
Ctr ThrRS	<i>Candida tropicalis</i> ATCC750	Yeast
Ctr VMA	<i>Candida tropicalis</i> (nucleus)	Yeast
Ctr-MYA3404 VMA	<i>Candida tropicalis</i> MYA-3404	Taxon: 294747
Ddi RPC2	<i>Dictyostelium discoideum</i> strain AX4 (nucleus)	Mycetozoa (a social amoeba)
Dhan GLT1	<i>Debaryomyces hansenii</i> CBS767	Fungi, Anamorph: <i>Candida famata</i> , taxon: 4959
Dhan VMA	<i>Debaryomyces hansenii</i> CBS767	Fungi, taxon: 284592
Eni PRP8	<i>Emericella nidulans</i> R20 (anamorph: <i>Aspergillus nidulans</i>)	taxon: 162425
Eni-FGSCA4 PRP8	<i>Emericella nidulans</i> (anamorph: <i>Aspergillus nidulans</i>) FGSC A4	Filamentous fungus, taxon: 162425
Fte RPB2 (RpoB)	<i>Floydella terrestris</i> , strain UTEX 1709	Green alga, chloroplast gene, taxon: 51328
Gth DnaB	<i>Guillardia theta</i> (plastid)	Cryptophyte Algae
HaV01 Pol	<i>Heterosigma akashiwo</i> virus 01	Algal virus, taxon: 97195, strain HaV01
Hca PRP8	<i>Histoplasma capsulatum</i> (anamorph: <i>Ajellomyces capsulatus</i>)	Fungi, human pathogen
IIV6 RIR1	Invertebrate iridescent virus 6	dsDNA eucaryotic virus, taxon: 176652
Kex-CBS379 VMA	<i>Kazachstania exigua</i> , formerly <i>Saccharomyces exiguus</i> , strain CBS379	Yeast, taxon: 34358
Kla-CBS683 VMA	<i>Kluyveromyces lactis</i> , strain CBS683	Yeast, taxon: 28985
Kla-IFO1267 VMA	<i>Kluyveromyces lactis</i> IFO1267	Fungi, taxon: 28985
Kla-NRRLY1140 VMA	<i>Kluyveromyces lactis</i> NRRL Y-1140	Fungi, taxon: 284590
Lel VMA	<i>Lodderomyces elongisporus</i>	Yeast
Mca-CBS113480 PRP8	<i>Microsporium canis</i> CBS 113480	Taxon: 554155

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Nau PRP8	<i>Neosartorya aurata</i> NRRL 4378	Fungus, taxon: 41051
Nfe-NRRL5534 PRP8	<i>Neosartorya fennelliae</i> NRRL 5534	Fungus, taxon: 41048
Nfi PRP8	<i>Neosartorya fischeri</i>	Fungi
Ngl-FR2163 PRP8	<i>Neosartorya glabra</i> FRR2163	Fungi, ATCC 16909, taxon: 41049
Ngl-FRR1833 PRP8	<i>Neosartorya glabra</i> FRR1833	Fungi, taxon: 41049, (preliminary identification) taxon: 41053
Nqu PRP8	<i>Neosartorya quadricincta</i> , strain NRRL 4175	
Nspi PRP8	<i>Neosartorya spinosa</i> FRR4595	Fungi, taxon: 36631
Pabr-Pb01 PRP8	<i>Paracoccidioides brasiliensis</i> Pb01	Taxon: 502779
Pabr-Pb03 PRP8	<i>Paracoccidioides brasiliensis</i> Pb03	Taxon: 482561
Pan CHS2	<i>Podospira anserina</i>	Fungi, Taxon 5145
Pan GLT1	<i>Podospira anserina</i>	Fungi, Taxon 5145
Pbl PRP8-a	<i>Phycomyces blakesleeana</i>	Zygomycete fungus, strain NRRL155
Pbl PRP8-b	<i>Phycomyces blakesleeana</i>	Zygomycete fungus, strain NRRL155
Pbr-Pb18 PRP8	<i>Paracoccidioides brasiliensis</i> Pb18	Fungi, taxon: 121759
Pch PRP8	<i>Penicillium chrysogenum</i>	Fungus, taxon: 5076
Pex PRP8	<i>Penicillium expansum</i>	Fungus, taxon: 27334
Pgu GLT1	<i>Pichia (Candida) guilliermondii</i>	Fungi, Taxon 294746
Pgu-alt GLT1	<i>Pichia (Candida) guilliermondii</i>	Fungi
Pno GLT1	<i>Phaeosphaeria nodorum</i> SN15	Fungi, taxon: 321614
Pno RPA2	<i>Phaeosphaeria nodorum</i> SN15	Fungi, taxon: 321614
Ppu DnaB	<i>Porphyra purpurea</i> (chloroplast)	Red Alga
Pst VMA	<i>Pichia stipitis</i> CBS 6054, taxon: 322104	Yeast
Ptr PRP8	<i>Pyrenophora tritici-repentis</i> Pt-1C-BF	Ascomycete fungus, taxon: 426418
Pvu PRP8	<i>Penicillium vulpinum</i> (formerly <i>P. claviforme</i>)	Fungus
Pye DnaB	<i>Porphyra yezoensis</i> chloroplast, cultivar U-51	Red alga, organelle = "plastid: chloroplast", "taxon: 2788
Sas RPB2	<i>Spiromyces spiralis</i> NRRL 22631	Zygomycete fungus, isolate = "AFTOL-ID 185", taxon: 68401
Sca-CBS4309 VMA	<i>Saccharomyces castellii</i> , strain CBS4309	Yeast, taxon: 27288
Sca-IFO1992 VMA	<i>Saccharomyces castellii</i> , strain IFO1992	Yeast, taxon: 27288
Scar VMA	<i>Saccharomyces cariocanus</i> , strain = "UFRJ 50791	Yeast, taxon: 114526
Sce VMA	<i>Saccharomyces cerevisiae</i> (nucleus)	Yeast, also in Sce strains OUT7163, OUT7045, OUT7163, IFO1992
Sce-DH1-1A VMA	<i>Saccharomyces cerevisiae</i> strain DH1-1A	Yeast, taxon: 173900, also in Sce strains OUT7900, OUT7903, OUT7112
Sce-JAY291 VMA	<i>Saccharomyces cerevisiae</i> JAY291	Taxon: 574961
Sce-OUT7091 VMA	<i>Saccharomyces cerevisiae</i> OUT7091	Yeast, taxon: 4932, also in Sce strains OUT7043, OUT7064
Sce-OUT7112 VMA	<i>Saccharomyces cerevisiae</i> OUT7112	Yeast, taxon: 4932, also in Sce strains OUT7900, OUT7903
Sce-YJM789 VMA	<i>Saccharomyces cerevisiae</i> strain YJM789	Yeast, taxon: 307796
Sda VMA	<i>Saccharomyces dairenensis</i> , strain CBS 421	Yeast, taxon: 27289, Also in Sda strain IFO0211
Sex-IFO1128 VMA	<i>Saccharomyces exiguus</i> , strain = "IFO1128"	Yeast, taxon: 34358
She RPB2 (RpoB)	<i>Stigeoclonium helveticum</i> , strain UTEX 441	Green alga, chloroplast gene, taxon: 55999
Sja VMA	<i>Schizosaccharomyces japonicus</i> yFS275	Ascomycete fungus, taxon: 402676
Spa VMA	<i>Saccharomyces pastorianus</i> IFO11023	Yeast, taxon: 27292
Spu PRP8	<i>Spizellomyces punctatus</i>	Chytrid fungus,
Sun VMA	<i>Saccharomyces unisporus</i> , strain CBS 398	Yeast, taxon: 27294
Tgl VMA	<i>Torulaspora globosa</i> , strain CBS 764	Yeast, taxon: 48254
Tpr VMA	<i>Torulaspora pretoriensis</i> , strain CBS 5080	Yeast, taxon: 35629

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Ure-1704 PRP8	<i>Uncinocarpus reesii</i>	Filamentous fungus
Vpo VMA	<i>Vanderwaltozyma polyspora</i> , formerly <i>Kluyveromyces polysporus</i> , strain CBS 2163	Yeast, taxon: 36033
WIV RIR1	<i>Wiseana</i> indescens virus	dsDNA eucaryotic virus, taxon: 68347
Zba VMA	<i>Zygosaccharomyces bailii</i> , strain CBS 685	Yeast, taxon: 4954
Zbi VMA	<i>Zygosaccharomyces bisporus</i> , strain CBS 702	Yeast, taxon: 4957
Zro VMA	<i>Zygosaccharomyces rouxii</i> , strain CBS 688	Yeast, taxon: 4956
Eubacteria		
AP-APSE1 dpol	<i>Acyrtosiphon pisum</i> secondary endosymbiot phage 1	Bacteriophage, taxon: 67571
AP-APSE2 dpol	Bacteriophage APSE-2, isolate = T5A	Bacteriophage of <i>Candidatus Hamiltonella defensa</i> , endosymbiot of <i>Acyrtosiphon pisum</i> , taxon: 340054
AP-APSE4 dpol	Bacteriophage of <i>Candidatus Hamiltonella defensa</i> strain 5ATac, endosymbiot of <i>Acyrtosiphon pisum</i>	Bacteriophage, taxon: 568990
AP-APSE5 dpol	Bacteriophage APSE-5	Bacteriophage of <i>Candidatus Hamiltonella defensa</i> , endosymbiot of <i>Uroleucon rudbeckiae</i> , taxon: 568991
AP-Aaphi23 MupF	Bacteriophage Aaphi23, <i>Haemophilus</i> phage Aaphi23	<i>Actinobacillus actinomycetemcomitans</i> Bacteriophage, taxon: 230158
Aae RIR2	<i>Aquifex aeolicus</i> strain VF5	Thermophilic chemolithoautotroph, taxon: 63363 taxon: 397945
Aave-AAC001	<i>Acidovorax avenae</i> subsp. <i>citulli</i>	taxon: 397945
Aave1721	AAC00-1	
Aave-AAC001 RIR1	<i>Acidovorax avenae</i> subsp. <i>citulli</i> AAC00-1	Taxon: 643561
Aave-ATCC19860	<i>Acidovorax avenae</i> subsp. <i>avenae</i> ATCC 19860	taxon: 405416
Aba Hyp-02185	<i>Acinetobacter baumannii</i> ACICU	taxon: 351607
Ace RIR1	<i>Acidothermus cellulolyticus</i> 11B	taxon: 187272
Aeh DnaB-1	<i>Alkalilimnicola ehrlichei</i> MLHE-1	taxon: 187272
Aeh DnaB-2	<i>Alkalilimnicola ehrlichei</i> MLHE-1	taxon: 187272
Aeh RIR1	<i>Alkalilimnicola ehrlichei</i> MLHE-1	Taxon: 683735
AgP-S1249 MupF	<i>Aggregatibacter</i> phage S1249	<i>Cyanobacterium</i> , taxon: 72020
Aha DnaE-c	<i>Aphanothece halophytica</i>	<i>Cyanobacterium</i> , taxon: 72020
Aha DnaE-n	<i>Aphanothece halophytica</i>	Taxon: 572477
Alvi-DSM180 GyrA	<i>Allochrochromatium vinosum</i> DSM 180	Probably prophage gene, taxon: 314275
Ama MADE823	phage uncharacterized protein [<i>Alteromonas macleodii</i> 'Deep ecotype']	Taxon: 513049
Amax-CS328 DnaX	<i>Arthrospira maxima</i> CS-328	<i>Cyanobacterium</i> , taxon: 75695
Aov DnaE-c	<i>Aphanizomenon ovalisporum</i>	<i>Cyanobacterium</i> , taxon: 75695
Aov DnaE-n	<i>Aphanizomenon ovalisporum</i>	Taxon: 118562, strain C1
Apl-C1 DnaX	<i>Arthrospira platensis</i>	taxon: 290399
Arsp-FB24 DnaB	<i>Arthrobacter</i> species FB24	<i>Cyanobacterium</i> , Nitrogen- fixing, taxon: 103690
Asp DnaE-c	<i>Anabaena</i> species PCC7120, (<i>Nostoc</i> sp. PCC7120)	<i>Cyanobacterium</i> , Nitrogen- fixing, taxon: 103690
Asp DnaE-n	<i>Anabaena</i> species PCC7120, (<i>Nostoc</i> sp. PCC7120)	<i>Cyanobacterium</i> , taxon: 240292
Ava DnaE-c	<i>Anabaena variabilis</i> ATCC29413	<i>Cyanobacterium</i> , taxon: 240292
Ava DnaE-n	<i>Anabaena variabilis</i> ATCC29413	taxon: 354
Avin RIR1 BIL	<i>Azotobacter vinelandii</i>	taxon: 406425
Bce-MCO3 DnaB	<i>Burkholderia cenocepacia</i> MC0-3	taxon: 350702
Bce-PC184 DnaB	<i>Burkholderia cenocepacia</i> PC184	Probably prophage gene, Taxon: 439292
Bse-MLS10 TerA	<i>Bacillus selenitireducens</i> MLS10	Prophage in <i>B. subtilis</i> M1918. taxon: 157928
BsuP-M1918 RIR1	<i>B. subtilis</i> M1918 (prophage)	<i>B. subtilis</i> taxon 1423. SPbeta c2 phage, taxon: 66797
BsuP-SPBc2 RIR1	<i>B. subtilis</i> strain 168 Sp beta c2 prophage	

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Bvi IcmO	<i>Burkholderia vietnamiensis</i> G4	plasmid = "pBVIE03". taxon: 269482
CP-P1201 Thy1	<i>Corynebacterium</i> phage P1201	lytic bacteriophage P1201 from <i>Corynebacterium glutamicum</i> NCHU 87078. Viruses; dsDNA viruses, taxon: 384848
Cag RIR1 Cau SpoVR	<i>Chlorochromatium aggregatum</i> <i>Chloroflexus aurantiacus</i> J-10-fl	Motile, phototrophic consortia Anoxygenic phototroph, taxon: 324602
CbP-C-St RNR	<i>Clostridium botulinum</i> phage C-St	Phage, specific _host = " <i>Clostridium botulinum</i> type C strain C-Stockholm, taxon: 12336
CbP-D1873 RNR	<i>Clostridium botulinum</i> phage D	Ssp. phage from <i>Clostridium botulinum</i> type D strain, 1873, taxon: 29342
Cbu-Dugway DnaB	<i>Coxiella burnetii</i> Dugway 5J108-111	Proteobacteria; Legionellales; taxon: 434922
Cbu-Goat DnaB	<i>Coxiella burnetii</i> 'MSU Goat Q177'	Proteobacteria; Legionellales; taxon: 360116
Cbu-RSA334 DnaB	<i>Coxiella burnetii</i> RSA 334	Proteobacteria; Legionellales; taxon: 360117
Cbu-RSA493 DnaB	<i>Coxiella burnetii</i> RSA 493	Proteobacteria; Legionellales; taxon: 227377
Cce Hyp1-Csp-2	<i>Cyanothece</i> sp. ATCC 51142	Marine unicellular diazotrophic <i>cyanobacterium</i> , taxon: 43989
Cch RIR1 Ccy Hyp1-Csp-1	<i>Chlorobium chlorochromatii</i> CaD3 <i>Cyanothece</i> sp. CCY0110	taxon: 340177 <i>Cyanobacterium</i> , taxon: 391612
Ccy Hyp1-Csp-2	<i>Cyanothece</i> sp. CCY0110	<i>Cyanobacterium</i> , taxon: 391612
Cfl-DSM20109 DnaB Chy RIR1	<i>Cellulomonas flavigena</i> DSM 20109 <i>Carboxydotherrmus</i> <i>hydrogenoformans</i> Z-2901	Taxon: 446466 Thermophile, taxon = 246194
Ckl PTerm	<i>Clostridium kluyveri</i> DSM 555	plasmid = "pCKL555A", taxon: 431943
Cra-CS505 DnaE-c Cra-CS505 DnaE-n Cra-CS505 GyrB	<i>Cylindrospermopsis raciborskii</i> CS-505 <i>Cylindrospermopsis raciborskii</i> CS-505 <i>Cylindrospermopsis raciborskii</i> CS-505	Taxon: 533240 Taxon: 533240 Taxon: 533240
Csp-CCY0110 DnaE-c Csp-CCY0110 DnaE-n	<i>Cyanothece</i> sp. CCY0110 <i>Cyanothece</i> sp. CCY0110	Taxon: 391612 Taxon: 391612
Csp-PCC7424 DnaE-c Csp-PCC7424 DnaE-n	<i>Cyanothece</i> sp. PCC 7424 <i>Cyanothece</i> sp. PCC7424	<i>Cyanobacterium</i> , taxon: 65393 <i>Cyanobacterium</i> , taxon: 65393
Csp-PCC7425 DnaB Csp-PCC7822 DnaE-n	<i>Cyanothece</i> sp. PCC 7425 <i>Cyanothece</i> sp. PCC 7822	Taxon: 395961 Taxon: 497965
Csp-PCC8801 DnaE-c Csp-PCC8801 DnaE-n	<i>Cyanothece</i> sp. PCC 8801 <i>Cyanothece</i> sp. PCC 8801	Taxon: 41431 Taxon: 41431
Cth ATPase BIL Cth-ATCC27405 TerA	<i>Clostridium thermocellum</i> <i>Clostridium thermocellum</i> ATCC27405	ATCC27405, taxon: 203119 Probable prophage, ATCC27405, taxon: 203119
Cth-DSM2360 TerA	<i>Clostridium thermocellum</i> DSM 2360	Probably prophage gene, Taxon: 572545
Cwa DnaB	<i>Crocospaera watsonii</i> WH 8501 (<i>Synechocystis</i> sp. WH 8501)	taxon: 165597
Cwa DnaE-c	<i>Crocospaera watsonii</i> WH 8501 (<i>Synechocystis</i> sp. WH 8501)	<i>Cyanobacterium</i> , taxon: 165597
Cwa DnaE-n	<i>Crocospaera watsonii</i> WH 8501 (<i>Synechocystis</i> sp. WH 8501)	<i>Cyanobacterium</i> , taxon: 165597
Cwa PEP	<i>Crocospaera watsonii</i> WH 8501 (<i>Synechocystis</i> sp. WH 8501)	taxon: 165597
Cwa RIR1	<i>Crocospaera watsonii</i> WH 8501 (<i>Synechocystis</i> sp. WH 8501)	taxon: 165597
Daud RIR1	<i>Candidatus Desulforudis audaxviator</i> MP104C	taxon: 477974
Dge DnaB	<i>Deinococcus geothermalis</i> DSM11300	Thermophilic, radiation resistant
Dha-DCB2 RIR1	<i>Desulfitobacterium hafniense</i> DCB-2	Anaerobic dehalogenating bacteria, taxon: 49338
Dha-Y51 RIR1	<i>Desulfitobacterium hafniense</i> Y51	Anaerobic dehalogenating bacteria, taxon: 138119

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Dpr-MLMSI RIR1	delta proteobacterium MLMS-1	Taxon: 262489
Dra RIR1	<i>Deinococcus radiodurans</i> R1, TIGR strain	Radiation resistant, taxon: 1299
Dra Snf2-c	<i>Deinococcus radiodurans</i> R1, TIGR strain	Radiation and DNA damage resistant, taxon: 1299
Dra Snf2-n	<i>Deinococcus radiodurans</i> R1, TIGR strain	Radiation and DNA damage resistant, taxon: 1299
Dra-ATCC13939 Snf2	<i>Deinococcus radiodurans</i> R1, ATCC13939/Brooks & Murray strain	Radiation and DNA damage resistant, taxon: 1299
Dth UDP GD	<i>Dictyoglomus thermophilum</i> H-6-12	strain = "H-6-12; ATCC 35947, taxon: 309799
Dvul ParB	<i>Desulfovibrio vulgaris</i> subsp. <i>vulgaris</i> DP4	taxon: 391774
EP-Min27 Primase	Enterobacteria phage Min27	bacteriophage of host = " <i>Escherichia coli</i> O157: H7 str. Min27"
Fal DnaB	<i>Frankia alni</i> ACN14a	Plant symbiot, taxon: 326424
Fsp-CcI3 RIR1	<i>Frankia</i> species CcI3	taxon: 106370
Gob DnaE	<i>Gemmata obscuriglobus</i> UQM2246	Taxon 114, TIGR genome strain, budding bacteria
Gob Hyp	<i>Gemmata obscuriglobus</i> UQM2246	Taxon 114, TIGR genome strain, budding bacteria
Gvi DnaB	<i>Gloeobacter violaceus</i> , PCC 7421	taxon: 33072
Gvi RIR1-1	<i>Gloeobacter violaceus</i> , PCC 7421	taxon: 33072
Gvi RIR1-2	<i>Gloeobacter violaceus</i> , PCC 7421	taxon: 33072
Hhal DnaB	<i>Halorhodospira halophila</i> SL1	taxon: 349124
Kfl-DSM17836 DnaB	<i>Kribbella flavida</i> DSM 17836	Taxon: 479435
Kra DnaB	<i>Kineococcus radiotolerans</i> SRS30216	Radiation resistant
LLP-KSY1 PolA	<i>Lactococcus</i> phage KSY1	Bacteriophage, taxon: 388452
LP-phiHSIC Helicase	<i>Listonella pelagia</i> phage phiHSIC	taxon: 310539, a pseudotemperate marine phage of <i>Listonella pelagia</i>
Lsp-PCC8106 GyrB	<i>Lyngbya</i> sp. PCC 8106	Taxon: 313612
MP-Be DnaB	Mycobacteriophage Bethlehem	Bacteriophage, taxon: 260121
MP-Be gp51	Mycobacteriophage Bethlehem	Bacteriophage, taxon: 260121
MP-Catera gp206	Mycobacteriophage Catera	Mycobacteriophage, taxon: 373404
MP-KBG gp53	<i>Mycobacterium</i> phage KBG	Taxon: 540066
MP-Mcjlw1 DnaB	Mycobacteriophage CJW1	Bacteriophage, taxon: 205869
MP-Omega DnaB	Mycobacteriophage Omega	Bacteriophage, taxon: 205879
MP-U2 gp50	Mycobacteriophage U2	Bacteriophage, taxon: 260120
Maer-NIES843 DnaB	<i>Microcystis aeruginosa</i> NIES-843	Bloom-forming toxic <i>cyanobacterium</i> , taxon: 449447
Maer-NIES843 DnaE-c	<i>Microcystis aeruginosa</i> NIES-843	Bloom-forming toxic <i>cyanobacterium</i> , taxon: 449447
Maer-NIES843 DnaE-n	<i>Microcystis aeruginosa</i> NIES-843	Bloom-forming toxic <i>cyanobacterium</i> , taxon: 449447
Mau-ATCC27029 GyrA	<i>Micromonospora aurantiaca</i> ATCC 27029	Taxon: 644283
Mav-104 DnaB	<i>Mycobacterium avium</i> 104	taxon: 243243
Mav-ATCC25291 DnaB	<i>Mycobacterium avium</i> subsp. <i>avium</i> ATCC 25291	Taxon: 553481
Mav-ATCC35712 DnaB	<i>Mycobacterium avium</i>	ATCC35712, taxon 1764
Mav-PT DnaB	<i>Mycobacterium avium</i> subsp. <i>paratuberculosis</i> str. k10	taxon: 262316
Mbo Pps1	<i>Mycobacterium bovis</i> subsp. <i>bovis</i> AF2122/97	strain = "AF2122/97", taxon: 233413
Mbo RecA	<i>Mycobacterium bovis</i> subsp. <i>bovis</i> AF2122/97	taxon: 233413
Mbo SufB (Mbo Pps1)	<i>Mycobacterium bovis</i> subsp. <i>bovis</i> AF2122/97	taxon: 233413
Mbo-1173P DnaB	<i>Mycobacterium bovis</i> BCG Pasteur 1173P	strain = BCG Pasteur 1173P2,, taxon: 410289
Mbo-AF2122 DnaB	<i>Mycobacterium bovis</i> subsp. <i>bovis</i> AF2122/97	strain = "AF2122/97", taxon: 233413
Mca MupF	<i>Methylococcus capsulatus</i> Bath, prophage MuMc02	prophage MuMc02, taxon: 243233
Mca RIR1	<i>Methylococcus capsulatus</i> Bath	taxon: 243233
Mch RecA	<i>Mycobacterium chitae</i>	IP14116003, taxon: 1792
Mcht-PCC7420 DnaE-1	<i>Microcoleus chthonoplastes</i> PCC7420	<i>Cyanobacterium</i> , taxon: 118168

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Mcht-PCC7420 DnaE-2c	<i>Microcoleus chthonoplastes</i> PCC7420	<i>Cyanobacterium</i> , taxon: 118168
Mcht-PCC7420 DnaE-2n	<i>Microcoleus chthonoplastes</i> PCC7420	<i>Cyanobacterium</i> , taxon: 118168
Mcht-PCC7420 GyrB	<i>Microcoleus chthonoplastes</i> PCC 7420	Taxon: 118168
Mcht-PCC7420 RIR1-1	<i>Microcoleus chthonoplastes</i> PCC 7420	Taxon: 118168
Mcht-PCC7420 RIR1-2	<i>Microcoleus chthonoplastes</i> PCC 7420	Taxon: 118168
Mex Helicase	<i>Methylobacterium extorquens</i> AM1	Alphaproteobacteria
Mex TrbC	<i>Methylobacterium extorquens</i> AM1	Alphaproteobacteria
Mfa RecA	<i>Mycobacterium fallax</i>	CITP8139, taxon: 1793
Mfl GyrA	<i>Mycobacterium flavescens</i> Fla0	taxon: 1776, reference #930991
Mfl RecA	<i>Mycobacterium flavescens</i> Fla0	strain = Fla0, taxon: 1776, ref. #930991
Mfl-ATCC14474 RecA	<i>Mycobacterium flavescens</i> , ATCC14474	strain = ATCC14474, taxon: 1776, ref #930991
Mfl-PYR-GCK DnaB	<i>Mycobacterium flavescens</i> PYR-GCK	taxon: 350054
Mga GyrA	<i>Mycobacterium gastri</i>	HP4389, taxon: 1777
Mga RecA	<i>Mycobacterium gastri</i>	HP4389, taxon: 1777
Mga SuFB (Mga Pps1)	<i>Mycobacterium gastri</i>	HP4389, taxon: 1777
Mgi-PYR-GCK DnaB	<i>Mycobacterium gilvum</i> PYR-GCK	taxon: 350054
Mgi-PYR-GCK GyrA	<i>Mycobacterium gilvum</i> PYR-GCK	taxon: 350054
Mgo GyrA	<i>Mycobacterium goodii</i>	taxon: 1778, reference number 930835
Min-1442 DnaB	<i>Mycobacterium intracellulare</i>	strain 1442, taxon: 1767
Min-ATCC13950	<i>Mycobacterium intracellulare</i> ATCC 13950	Taxon: 487521
GyrA		
Mkas GyrA	<i>Mycobacterium kansasii</i>	taxon: 1768
Mkas-ATCC12478	<i>Mycobacterium kansasii</i> ATCC 12478	Taxon: 557599
GyrA		
Mle-Br4923 GyrA	<i>Mycobacterium leprae</i> Br4923	Taxon: 561304
Mle-TN DnaB	<i>Mycobacterium leprae</i> , strain TN	Human pathogen, taxon: 1769
Mle-TN GyrA	<i>Mycobacterium leprae</i> TN	Human pathogen, STRAIN = TN, taxon: 1769
Mle-TN RecA	<i>Mycobacterium leprae</i> , strain TN	Human pathogen, taxon: 1769
Mle-TN SuFB (Mle Pps1)	<i>Mycobacterium leprae</i>	Human pathogen, taxon: 1769
Mma GyrA	<i>Mycobacterium mageritense</i>	taxon: 1780
Mmag Magn8951 BIL	<i>Magnetospirillum magnetotacticum</i> MS-1	Gram negative, taxon: 272627
Msh RecA	<i>Mycobacterium shimodei</i>	ATCC27962, taxon: 29313
Msm DnaB-1	<i>Mycobacterium smegmatis</i> MC2 155	MC2 155, taxon: 246196
Msm DnaB-2	<i>Mycobacterium smegmatis</i> MC2 155	MC2 155, taxon: 246196
Msp-KMS DnaB	<i>Mycobacterium species</i> KMS	taxon: 189918
Msp-KMS GyrA	<i>Mycobacterium species</i> KMS	taxon: 189918
Msp-MCS DnaB	<i>Mycobacterium species</i> MCS	taxon: 164756
Msp-MCS GyrA	<i>Mycobacterium species</i> MCS	taxon: 164756
Mthe RecA	<i>Mycobacterium thermoresistibile</i>	ATCC19527, taxon: 1797
Mtu SuFB (Mtu Pps1)	<i>Mycobacterium tuberculosis</i> strains H37Rv & CDC1551	Human pathogen, taxon: 83332
Mtu-C RecA	<i>Mycobacterium tuberculosis</i> C	Taxon: 348776
Mtu-CDC1551 DnaB	<i>Mycobacterium tuberculosis</i> , CDC1551	Human pathogen, taxon: 83332
Mtu-CPHL RecA	<i>Mycobacterium tuberculosis</i> CPHL_A	Taxon: 611303
Mtu-Canetti RecA	<i>Mycobacterium tuberculosis</i> / strain = "Canetti"	Taxon: 1773
Mtu-EAS054 RecA	<i>Mycobacterium tuberculosis</i> EAS054	Taxon: 520140
Mtu-F11 DnaB	<i>Mycobacterium tuberculosis</i> , strain F11	taxon: 336982
Mtu-H37Ra DnaB	<i>Mycobacterium tuberculosis</i> H37Ra	ATCC 25177, taxon: 419947
Mtu-H37Rv DnaB	<i>Mycobacterium tuberculosis</i> H37Rv	Human pathogen, taxon: 83332
Mtu-H37Rv RecA	<i>Mycobacterium tuberculosis</i> H37Rv, Also CDC1551	Human pathogen, taxon: 83332
Mtu-Haarlem DnaB	<i>Mycobacterium tuberculosis</i> str. Haarlem	Taxon: 395095
Mtu-K85 RecA	<i>Mycobacterium tuberculosis</i> K85	Taxon: 611304
Mtu-R604 RecA-n	<i>Mycobacterium tuberculosis</i> '98-R604 INH-RIF-EM'	Taxon: 555461

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Mtu-So93 RecA	<i>Mycobacterium tuberculosis</i> So93/sub__species = "Canetti"	Human pathogen, taxon: 1773
Mtu-T17 RecA-c	<i>Mycobacterium tuberculosis</i> T17	Taxon: 537210
Mtu-T17 RecA-n	<i>Mycobacterium tuberculosis</i> T17	Taxon: 537210
Mtu-T46 RecA	<i>Mycobacterium tuberculosis</i> T46	Taxon: 611302
Mtu-T85 RecA	<i>Mycobacterium tuberculosis</i> T85	Taxon: 520141
Mtu-T92 RecA	<i>Mycobacterium tuberculosis</i> T92	Taxon: 515617
Mvan DnaB	<i>Mycobacterium vanbaalenii</i> PYR-1	taxon: 350058
Mvan GyrA	<i>Mycobacterium vanbaalenii</i> PYR-1	taxon: 350058
Mxa RAD25	<i>Myxococcus xanthus</i> DK1622	Deltaproteobacteria
Mxe GyrA	<i>Mycobacterium xenopi</i> strain IMM5024	taxon: 1789
Naz-0708 RIR1-1	<i>Nostoc azollae</i> 0708	Taxon: 551115
Naz-0708 RIR1-2	<i>Nostoc azollae</i> 0708	Taxon: 551115
Nfa DnaB	<i>Nocardia farcinica</i> IFM 10152	taxon: 247156
Nfa Nfa15250	<i>Nocardia farcinica</i> IFM 10152	taxon: 247156
Nfa RIR1	<i>Nocardia farcinica</i> IFM 10152	taxon: 247156
Nosp-CCY9414 DnaE-n	<i>Nodularia spumigena</i> CCY9414	Taxon: 313624
Npu DnaB	<i>Nostoc punctiforme</i>	<i>Cyanobacterium</i> , taxon: 63737
Npu GyrB	<i>Nostoc punctiforme</i>	<i>Cyanobacterium</i> , taxon: 63737
Npu-PCC73102 DnaE-c	<i>Nostoc punctiforme</i> PCC73102	<i>Cyanobacterium</i> , taxon: 63737, ATCC29133
Npu-PCC73102 DnaE-n	<i>Nostoc punctiforme</i> PCC73102	<i>Cyanobacterium</i> , taxon: 63737, ATCC29133
Nsp-JS614 DnaB	<i>Nocardioides</i> species JS614	taxon: 196162
Nsp-JS614 TOPRIM	<i>Nocardioides</i> species JS614	taxon: 196162
Nsp-PCC7120 DnaB	<i>Nostoc</i> species PCC7120, (<i>Anabaena</i> sp. PCC7120)	<i>Cyanobacterium</i> , Nitrogen-fixing, taxon: 103690
Nsp-PCC7120 DnaE-c	<i>Nostoc</i> species PCC7120, (<i>Anabaena</i> sp. PCC7120)	<i>Cyanobacterium</i> , Nitrogen-fixing, taxon: 103690
Nsp-PCC7120 DnaE-n	<i>Nostoc</i> species PCC7120, (<i>Anabaena</i> sp. PCC7120)	<i>Cyanobacterium</i> , Nitrogen-fixing, taxon: 103690
Nsp-PCC7120 RIR1	<i>Nostoc</i> species PCC7120, (<i>Anabaena</i> sp. PCC7120)	<i>Cyanobacterium</i> , Nitrogen-fixing, taxon: 103690
Oli DnaE-c	<i>Oscillatoria limnetica</i> str. 'Solar Lake'	<i>Cyanobacterium</i> , taxon: 262926
Oli DnaE-n	<i>Oscillatoria limnetica</i> str. 'Solar Lake'	<i>Cyanobacterium</i> , taxon: 262926
PP-PhiEL Helicase	<i>Pseudomonas aeruginosa</i> phage phiEL	Phage infects <i>Pseudomonas aeruginosa</i> , taxon: 273133
PP-PhiEL ORF11	<i>Pseudomonas aeruginosa</i> phage phiEL	phage infects <i>Pseudomonas aeruginosa</i> , taxon: 273133
PP-PhiEL ORF39	<i>Pseudomonas aeruginosa</i> phage phiEL	Phage infects <i>Pseudomonas aeruginosa</i> , taxon: 273133
PP-PhiEL ORF40	<i>Pseudomonas aeruginosa</i> phage phiEL	phage infects <i>Pseudomonas aeruginosa</i> , taxon: 273133
Pfi Fha BIL	<i>Pseudomonas fluorescens</i> Pf-5	Plant commensal organism, taxon: 220664
Plut RIR1	<i>Pelodictyon luteolum</i> DSM 273	Green sulfur bacteria, Taxon 319225
Pma-EXH1 GyrA	<i>Persephonella marina</i> EX-H1	Taxon: 123214
Pma-ExH1 DnaE	<i>Persephonella marina</i> EX-H1	Taxon: 123214
Pna RIR1	<i>Polaromonas naphthalenivorans</i> CJ2	taxon: 365044
Pnuc DnaB	<i>Polynucleobacter</i> sp. QLW-P1DMWA-1	taxon: 312153
Posp-JS666 DnaB	<i>Polaromonas</i> species JS666	taxon: 296591
Posp-JS666 RIR1	<i>Polaromonas</i> species JS666	taxon: 296591
Pssp-A1-1 Fha	<i>Pseudomonas</i> species A1-1	
Psy Fha	<i>Pseudomonas syringae</i> pv. tomato str. DC3000	Plant (tomato) pathogen, taxon: 223283
Rbr-D9 GyrB	<i>Raphidiopsis brookii</i> D9	Taxon: 533247
Rce RIR1	<i>Rhodospirillum centenum</i> SW	taxon: 414684, ATCC 51521
Rer-SK121 DnaB	<i>Rhodococcus erythropolis</i> SK121	Taxon: 596309
Rma DnaB	<i>Rhodothermus marinus</i>	Thermophile, taxon: 29549
Rma-DSM4252 DnaB	<i>Rhodothermus marinus</i> DSM 4252	Taxon: 518766
Rma-DSM4252 DnaE	<i>Rhodothermus marinus</i> DSM 4252	Thermophile, taxon: 518766
Rsp RIR1	<i>Roseovarius</i> species 217	taxon: 314264
SaP-SETP12 dpol	<i>Salmonella</i> phage SETP12	Phage, taxon: 424946
SaP-SETP3 Helicase	<i>Salmonella</i> phage SETP3	Phage, taxon: 424944
SaP-SETP3 dpol	<i>Salmonella</i> phage SETP3	Phage, taxon: 424944
SaP-SETP5 dpol	<i>Salmonella</i> phage SETP5	Phage, taxon: 424945
Sare DnaB	<i>Salinispora arenicola</i> CNS-205	taxon: 391037

TABLE 1-continued

[illegible]

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Ter Ndse-1	<i>Trichodesmium erythraeum</i> IMS101	<i>Cyanobacterium</i> , taxon: 203124
Ter Ndse-2	<i>Trichodesmium erythraeum</i> IMS101	<i>Cyanobacterium</i> , taxon: 203124
Ter RIR1-1	<i>Trichodesmium erythraeum</i> IMS101	<i>Cyanobacterium</i> , taxon: 203124
Ter RIR1-2	<i>Trichodesmium erythraeum</i> IMS101	<i>Cyanobacterium</i> , taxon: 203124
Ter RIR1-3	<i>Trichodesmium erythraeum</i> IMS101	<i>Cyanobacterium</i> , taxon: 203124
Ter RIR1-4	<i>Trichodesmium erythraeum</i> IMS101	<i>Cyanobacterium</i> , taxon: 203124
Ter Snf2	<i>Trichodesmium erythraeum</i> IMS101	<i>Cyanobacterium</i> , taxon: 203124
Ter ThyX	<i>Trichodesmium erythraeum</i> IMS101	<i>Cyanobacterium</i> , taxon: 203124
Tfus RecA-1	<i>Thermobifida fusca</i> YX	Thermophile, taxon: 269800
Tfus RecA-2	<i>Thermobifida fusca</i> YX	Thermophile, taxon: 269800
Tfus Tfu2914	<i>Thermobifida fusca</i> YX	Thermophile, taxon: 269800
Thsp-K90 RIR1	<i>Thioalkalivibrio</i> sp. K90mix	Taxon: 396595
Tth-DSM571 RIR1	<i>Thermoanaerobacterium thermosaccharolyticum</i> DSM 571	Taxon: 580327
Tth-HB27 DnaE-1	<i>Thermus thermophilus</i> HB27	thermophile, taxon: 262724
Tth-HB27 DnaE-2	<i>Thermus thermophilus</i> HB27	thermophile, taxon: 262724
Tth-HB27 RIR1-1	<i>Thermus thermophilus</i> HB27	thermophile, taxon: 262724
Tth-HB27 RIR1-2	<i>Thermus thermophilus</i> HB27	thermophile, taxon: 262724
Tth-HB8 DnaE-1	<i>Thermus thermophilus</i> HB8	thermophile, taxon: 300852
Tth-HB8 DnaE-2	<i>Thermus thermophilus</i> HB8	thermophile, taxon: 300852
Tth-HB8 RIR1-1	<i>Thermus thermophilus</i> HB8	thermophile, taxon: 300852
Tth-HB8 RIR1-2	<i>Thermus thermophilus</i> HB8	thermophile, taxon: 300852
Tvu DnaE-c	<i>Thermosynechococcus vulcanus</i>	<i>Cyanobacterium</i> , taxon: 32053
Tvu DnaE-n	<i>Thermosynechococcus vulcanus</i>	<i>Cyanobacterium</i> , taxon: 32053
Tye RNR-1	<i>Thermodesulfobivrio yellowstonii</i> DSM 11347	taxon: 289376
Tye RNR-2	<i>Thermodesulfobivrio yellowstonii</i> DSM 11347	taxon: 289376
Archaea		
Ape APE0745	<i>Aeropyrum pernix</i> K1	Thermophile, taxon: 56636
Cme-boo Pol-II	<i>Candidatus Methanoregula boonei</i> 6A8	taxon: 456442
Fac-Fer1 RIR1	<i>Ferroplasma acidarmanus</i> , taxon: 97393 and taxon 261390	strain Fer1, eats iron
Fac-Fer1 SufB (Fac Pps1)	<i>Ferroplasma acidarmanus</i>	strain fer1, eats iron, taxon: 97393
Fac-TypeI RIR1	<i>Ferroplasma acidarmanus</i> type I,	Eats iron, taxon 261390
Fac-typeI SufB (Fac Pps1)	<i>Ferroplasma acidarmanus</i>	Eats iron, taxon: 261390
Hma CDC21	<i>Haloarcula marismortui</i> ATCC 43049	taxon: 272569,
Hma Pol-II	<i>Haloarcula marismortui</i> ATCC 43049	taxon: 272569,
Hma PolB	<i>Haloarcula marismortui</i> ATCC 43049	taxon: 272569,
Hma TopA	<i>Haloarcula marismortui</i> ATCC 43049	taxon: 272569
Hmu-DSM12286 MCM	<i>Halomicrobium mukohataei</i> DSM 12286	taxon: 485914 (<i>Halobacteria</i>)
Hmu-DSM12286 PolB	<i>Halomicrobium mukohataei</i> DSM 12286	Taxon: 485914
Hsa-R1 MCM	<i>Halobacterium salinarum</i> R-1	Halophile, taxon: 478009, strain = "R1; DSM 671"
Hsp-NRC1 CDC21	<i>Halobacterium</i> species NRC-1	Halophile, taxon: 64091
Hsp-NRC1 Pol-II	<i>Halobacterium salinarum</i> NRC-1	Halophile, taxon: 64091
Hut MCM-2	<i>Halorhabdus utahensis</i> DSM 12940	taxon: 519442
Hut-DSM12940 MCM-1	<i>Halorhabdus utahensis</i> DSM 12940	taxon: 519442
Hvo PolB	<i>Haloferax volcanii</i> DS70	taxon: 2246
Hwa GyrB	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa MCM-1	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa MCM-2	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa MCM-3	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Hwa MCM-4	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa Pol-II-1	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa Pol-II-2	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa PolB-1	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa PolB-2	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa PolB-3	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa RCF	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa RIR1-1	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa RIR1-2	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa Top6B	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Hwa rPol A"	<i>Haloquadratum walsbyi</i> DSM 16790	Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001
Maeo Pol-II	<i>Methanococcus aeolicus</i> Nankai-3	taxon: 419665
Maeo RFC	<i>Methanococcus aeolicus</i> Nankai-3	taxon: 419665
Maeo RNR	<i>Methanococcus aeolicus</i> Nankai-3	taxon: 419665
Maeo-N3 Helicase	<i>Methanococcus aeolicus</i> Nankai-3	taxon: 419665
Maeo-N3 RtcB	<i>Methanococcus aeolicus</i> Nankai-3	taxon: 419665
Maeo-N3 UDP GD	<i>Methanococcus aeolicus</i> Nankai-3	taxon: 419665
Mein-ME PEP	<i>Methanocaldococcus infernus</i> ME	thermophile, Taxon: 573063
Mein-ME RFC	<i>Methanocaldococcus infernus</i> ME	Taxon: 573063
Memar MCM2	<i>Methanoculleus marisnigri</i> JR1	taxon: 368407
Memar Pol-II	<i>Methanoculleus marisnigri</i> JR1	taxon: 368407
Mesp-FS406 PolB-1	<i>Methanocaldococcus</i> sp. FS406-22	Taxon: 644281
Mesp-FS406 PolB-2	<i>Methanocaldococcus</i> sp. FS406-22	Taxon: 644281
Mesp-FS406 PolB-3	<i>Methanocaldococcus</i> sp. FS406-22	Taxon: 644281
Mesp-FS406-22 LHR	<i>Methanocaldococcus</i> sp. FS406-22	Taxon: 644281
Mfe-AG86 Pol-1	<i>Methanocaldococcus fervens</i> AG86	Taxon: 573064
Mfe-AG86 Pol-2	<i>Methanocaldococcus fervens</i> AG86	Taxon: 573064
Mhu Pol-II	<i>Methanospirillum hungatei</i> JF-1	taxon 323259
Mja GF-6P	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja Helicase	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja Hyp-1	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja IF2	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja KlbA	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja PEP	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja Pol-1	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Mja Pol-2	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja RFC-1	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja RFC-2	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja RFC-3	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja RNR-1	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja RNR-2	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja RtcB (Mja Hyp-2)	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja TFIIB	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja UDP GD	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja r-Gyr	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja rPol A'	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mja rPol A''	<i>Methanococcus jannaschii</i> (<i>Methanocaldococcus jannaschii</i> DSM 2661)	Thermophile, DSM 2661, taxon: 2190
Mka CDC48	<i>Methanopyrus kandleri</i> AV19	Thermophile, taxon: 190192
Mka EF2	<i>Methanopyrus kandleri</i> AV19	Thermophile, taxon: 190192
Mka RFC	<i>Methanopyrus kandleri</i> AV19	Thermophile, taxon: 190192
Mka RtcB	<i>Methanopyrus kandleri</i> AV19	Thermophile, taxon: 190192
Mka VabB	<i>Methanopyrus kandleri</i> AV19	Thermophile, taxon: 190192
Mth RIR1	<i>Methanothermobacter</i> <i>thermautotrophicus</i> (<i>Methanobacterium</i> <i>thermautotrophicum</i>)	Thermophile, delta H strain
Mvu-M7 Helicase	<i>Methanocaldococcus vulcanius</i> M7	Taxon: 579137
Mvu-M7 Pol-1	<i>Methanocaldococcus vulcanius</i> M7	Taxon: 579137
Mvu-M7 Pol-2	<i>Methanocaldococcus vulcanius</i> M7	Taxon: 579137
Mvu-M7 Pol-3	<i>Methanocaldococcus vulcanius</i> M7	Taxon: 579137
Mvu-M7 UDP GD	<i>Methanocaldococcus vulcanius</i> M7	Taxon: 579137
Neq Pol-c	<i>Nanoarchaeum equitans</i> Kin4-M	Thermophile, taxon: 228908
Neq Pol-n	<i>Nanoarchaeum equitans</i> Kin4-M	Thermophile, taxon: 228908
Nma-ATCC43099 MCM	<i>Natrialba magadii</i> ATCC 43099	Taxon: 547559
Nma-ATCC43099 PolB-1	<i>Natrialba magadii</i> ATCC 43099	Taxon: 547559
Nma-ATCC43099 PolB-2	<i>Natrialba magadii</i> ATCC 43099	Taxon: 547559
Nph CDC21	<i>Natronomonas pharaonis</i> DSM 2160	taxon: 348780
Nph PolB-1	<i>Natronomonas pharaonis</i> DSM 2160	taxon: 348780
Nph PolB-2	<i>Natronomonas pharaonis</i> DSM 2160	taxon: 348780
Nph rPol A''	<i>Natronomonas pharaonis</i> DSM 2160	taxon: 348780
Pab CDC21-1	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab CDC21-2	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab IF2	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab KlbA	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab Lon	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Pab Moaa	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab Pol-II	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab RFC-1	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab RFC-2	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab RIR1-1	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab RIR1-2	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab RIR1-3	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab RtcB (Pab Hyp-2)	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Pab VMA	<i>Pyrococcus abyssi</i>	Thermophile, strain Orsay, taxon: 29292
Par RIR1	<i>Pyrobaculum arsenaticum</i> DSM 13514	taxon: 340102
Pfu CDC21	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pfu IF2	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pfu KlbA	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pfu Lon	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pfu RFC	<i>Pyrococcus furiosus</i>	Thermophile, DSM3638, taxon: 186497
Pfu RIR1-1	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pfu RIR1-2	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pfu RtcB (Pfu Hyp-2)	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pfu TopA	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pfu VMA	<i>Pyrococcus furiosus</i>	Thermophile, taxon: 186497, DSM3638
Pho CDC21-1	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho CDC21-2	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho IF2	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho KlbA	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho LHR	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho Lon	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho Pol I	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho Pol-II	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho RFC	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho RIR1	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho RadA	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho RtcB (Pho Hyp-2)	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho VMA	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Pho r-Gyr	<i>Pyrococcus horikoshii</i> OT3	Thermophile, taxon: 53953
Psp-GBD Pol	<i>Pyrococcus</i> species GB-D	Thermophile
Pto VMA	<i>Picrophilus torridus</i> , DSM 9790	DSM 9790, taxon: 263820, Thermoacidophile
Smar 1471	<i>Staphylothermus marinus</i> F1	taxon: 399550
Smar MCM2	<i>Staphylothermus marinus</i> F1	taxon: 399550
Tac-ATCC25905 VMA	<i>Thermoplasma acidophilum</i> , ATCC 25905	Thermophile, taxon: 2303
Tac-DSM1728 VMA	<i>Thermoplasma acidophilum</i> , DSM1728	Thermophile, taxon: 2303
Tag Pol-1 (Tsp-TY Pol-1)	<i>Thermococcus aggregans</i>	Thermophile, taxon: 110163
Tag Pol-2 (Tsp-TY Pol-2)	<i>Thermococcus aggregans</i>	Thermophile, taxon: 110163
Tag Pol-3 (Tsp-TY Pol-3)	<i>Thermococcus aggregans</i>	Thermophile, taxon: 110163
Tba Pol-II	<i>Thermococcus barophilus</i> MP	taxon: 391623
Tfu Pol-1	<i>Thermococcus fumicolans</i>	Thermophile, taxon: 46540
Tfu Pol-2	<i>Thermococcus fumicolans</i>	Thermophile, taxon: 46540
Thy Pol-1	<i>Thermococcus hydrothermalis</i>	Thermophile, taxon: 46539
Thy Pol-2	<i>Thermococcus hydrothermalis</i>	Thermophile, taxon: 46539
Tko CDC21-1	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko CDC21-2	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014

TABLE 1-continued

Naturally Occurring Inteins		
Intein Name	Organism Name	Organism Description
Tko Helicase	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko IF2	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko KlbA	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko LHR	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko Pol-1 (Pko Pol-1)	<i>Pyrococcus/Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko Pol-2 (Pko Pol-2)	<i>Pyrococcus/Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko Pol-II	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko RFC	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko RIR1-1	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko RIR1-2	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko RadA	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko TopA	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tko r-Gyr	<i>Thermococcus kodakaraensis</i> KOD1	Thermophile, taxon: 69014
Tli Pol-1	<i>Thermococcus litoralis</i>	Thermophile, taxon: 2265
Tli Pol-2	<i>Thermococcus litoralis</i>	Thermophile, taxon: 2265
Tma Pol	<i>Thermococcus marinus</i>	taxon: 187879
Ton-NA1 LHR	<i>Thermococcus onnurineus</i> NA1	Taxon: 523850
Ton-NA1 Pol	<i>Thermococcus onnurineus</i> NA1	taxon: 342948
Tpe Pol	<i>Thermococcus peptonophilus</i> strain SM2	taxon: 32644
Tsi-MM739 Lon	<i>Thermococcus sibiricus</i> MM 739	Thermophile, Taxon: 604354
Tsi-MM739 Pol-1	<i>Thermococcus sibiricus</i> MM 739	Taxon: 604354
Tsi-MM739 Pol-2	<i>Thermococcus sibiricus</i> MM 739	Taxon: 604354
Tsi-MM739 RFC	<i>Thermococcus sibiricus</i> MM 739	Taxon: 604354
Tsp AM4 RtcB	<i>Thermococcus</i> sp. AM4	Taxon: 246969
Tsp-AM4 LHR	<i>Thermococcus</i> sp. AM4	Taxon: 246969
Tsp-AM4 Lon	<i>Thermococcus</i> sp. AM4	Taxon: 246969
Tsp-AM4 RIR1	<i>Thermococcus</i> sp. AM4	Taxon: 246969
Tsp-GE8 Pol-1	<i>Thermococcus</i> species GE8	Thermophile, taxon: 105583
Tsp-GE8 Pol-2	<i>Thermococcus</i> species GE8	Thermophile, taxon: 105583
Tsp-GT Pol-1	<i>Thermococcus</i> species GT	taxon: 370106
Tsp-GT Pol-2	<i>Thermococcus</i> species GT	taxon: 370106
Tsp-UGL-20P Pol	<i>Thermococcus</i> sp. OGL-20P	taxon: 277988
Tthi Pol	<i>Thermococcus thioreducens</i>	Hyperthermophile
Tvo VMA	<i>Thermoplasma volcanium</i> GSS1	Thermophile, taxon: 50339
Tzi Pol	<i>Thermococcus zilligii</i>	taxon: 54076
Unc-ERS PFL	uncultured archaeon Gzfos13E1	isolation_source = "Eel River sediment", clone = "GZfos13E1", taxon: 285397
Unc-ERS RIR1	uncultured archaeon GZfos9C4	isolation_source = "Eel River sediment", taxon: 285366, clone = "GZfos9C4"
Unc-ERS RNR	uncultured archaeon GZfos10C7	isolation_source = "Eel River sediment", clone = "GZfos10C7", taxon: 285400
Unc-MetRFS MCM2	uncultured archaeon (Rice Cluster I)	Enriched methanogenic consortium from rice field soil, taxon: 198240

[0069] The split inteins of the disclosed compositions or that can be used in the disclosed methods can be modified, or mutated, inteins. A modified intein can comprise modifications to the N-terminal intein segment, the C-terminal intein segment, or both. The modifications can include additional amino acids at the N-terminus the C-terminus of either portion of the split intein, or can be within the either portion of the split intein. Table 2 shows a list of amino acids, their abbreviations, polarity, and charge.

TABLE 2

List of Amino Acids				
Amino Acid	3-Letter Code	1-Letter Code	Polarity	Charge
Alanine	Ala	A	nonpolar	neutral
Arginine	Arg	R	Basic	positive
Asparagine	Asn	N	polar	neutral
Aspartic acid	Asp	D	acidic	negative
Cysteine	Cys	C	nonpolar	neutral
Glutamic acid	Glu	E	acidic	negative
Glutamine	Gln	Q	polar	neutral
Glycine	Gly	G	nonpolar	neutral
Histidine	His	H	Basic	Positive (10%) Neutral (90%)
Isoleucine	Ile	I	nonpolar	neutral
Leucine	Leu	L	nonpolar	neutral
Lysine	Lys	K	Basic	positive
Methionine	Met	M	nonpolar	neutral
Phenylalanine	Phe	F	nonpolar	neutral
Proline	Pro	P	nonpolar	neutral
Serine	Ser	S	polar	neutral
Threonine	Thr	T	polar	neutral
Tryptophan	Trp	W	nonpolar	neutral
Tyrosine	Tyr	Y	polar	neutral
Valine	Val	V	nonpolar	neutral

[0070] The N-intein of the invention may be coupled to solid phase, such as a membrane, fiber, particle, bead or chip. The solid phase may be a chromatography resin of natural or synthetic origin, such as a natural or synthetic resin, preferably a polysaccharide such as agarose. The solid phase, such as a chromatography resin, may be provided with embedded magnetic particles. In another embodiment the solid phase is a non-diffusion limited resin/fibrous material.

[0071] In this case the solid phase may be formed from one or more polymeric nanofibre substrates, such as electrospun polymer nanofibres. Polymer nanofibres for use in the present invention typically have mean diameters from 10 nm to 1000 nm. The length of polymer nanofibres is not particularly limited. The polymer nanofibres can suitably be monofilament nanofibres and may e.g. have a circular, ellipsoidal or essentially circular/ellipsoidal cross section. Typically, the one or more polymer nanofibres are provided in the form of one or more non-woven sheets, each comprising one or more polymer nanofibers. A non-woven sheet comprising one or more polymer nanofibres is a mat of said one or more polymer nanofibres with each nanofibre oriented essentially randomly, i.e. it has not been fabricated so that the nanofibre or nanofibres adopts a particular pattern. Non-woven sheets typically have area densities from 1 to 40 g/m². Non-woven sheets typically have a thickness from 5 to 120 μ m. The polymer should be a polymer suitable for use

as a chromatography medium, i.e. an adsorbent, in a chromatography method. Suitable polymers include polyamides such as nylon, polyacrylic acid, polymethacrylic acid, polyacrylonitrile, polystyrene, polysulfones e.g. polyether-sulfone (PES), polycaprolactone, collagen, chitosan, polyethylene oxide, agarose, agarose acetate, cellulose, cellulose acetate, and combinations thereof.

[0072] The N-intein according to the invention may be immobilized on a solid support in a very high degree, 0.2-2 μ mole/ml N-intein is coupled per ml resin (swollen gel).

[0073] The N-intein according to the invention may be coupled to the solid phase via a Lys-tail, comprising one or more Lys, such as at least two, on the C-terminal. Alternatively, the N-intein is coupled to the solid phase via a Cys-tail on the C-terminal.

C-Intein Protein Variants

[0074] Preferably the invention also provides a C-intein comprising a split intein C-intein sequence or engineered variants thereof.

[0075] It will be appreciated that selection of the N-intein and C-intein can be from the same wild type split intein (e.g., both from Npu, or a variant of either the N- or C-intein, or alternatively can be selected from different wild type split inteins or the consensus split intein sequences, as it has been discovered that the affinity of a N-fragment for a different C-fragment (e.g., Npu N-fragment or variant thereof with Ssp C-fragment or variant thereof) still maintains sufficient binding affinity for use in the disclosed methods.

Split Intein Systems

[0076] Preferably, the invention provides a split intein system for affinity purification of a protein of interest (POI), comprising a N-intein and C-intein as described above.

[0077] Preferably the N-intein is attached to a solid phase and the C-intein is co-expressed with the POI and used as a tag for affinity purification of the POI. Vice versa is also possible, ie attaching the C-intein to a solid phase and using the N-intein as a tag, but the former is preferred.

[0078] In one embodiment the C-intein and an additional tag is co-expressed with the POI. The additional tag may be any conventional chromatography tag, such as an IEX tag or an affinity tag.

Methods of Purifying a Protein of Interest (POI)

[0079] The invention relates to a method for purification of a protein of interest (POI), using the split intein system according to the invention, comprising association of the C-intein and N-intein at neutral pH, such as 6-8, and in the presence of divalent cations (which impairs spontaneous cleavage); washing said solid phase in the presence of divalent cations; addition of a chelator to allow spontaneous cleavage between C-intein and POI; collection of tagless POI.

[0080] This protocol is suitable for protein non-sensitive for Zn. The advantages are long contact times are allowed with the resin and addition of large sample volume. Sample loading could be made for long times, such as up to 1.5 hours.

[0081] According to the invention more than 30% yield, preferably 50%, most preferably more than 80% of POI is achieved in less than 4 hours cleavage.

[0082] The invention enables a high ligand density when the N-intein is immobilized to a solid phase. Preferably the N-intein is attached to a chromatography resin, such as agarose or any other suitable resin for protein purification. According to the invention it is possible to achieve a static binding capacity of 0.2-2 $\mu\text{mole/ml}$ C-intein bound POI per settled ml resin.

Affinity Tags

[0083] The invention also relates to a method for purification of a protein of interest (POI), comprising the following steps: co-expressing a POI with a C-intein according to the invention and an additional tag; binding said additional tag to its binding partner on a solid phase; cleaving off the POI and the C-intein; binding said C-intein to an N-intein attached to a solid phase at neutral pH and cleaving off said bound C-intein and N-intein from said POI; and re-generating said solid phase under alkaline conditions, such as 0.5M NaOH. The purpose of this twin tag: increased purity (enables dual affinity purification), solubility, detectability.

[0084] Affinity tags can be peptide or protein sequences cloned in frame with protein coding sequences that change the protein's behavior. Affinity tags can be appended to the N- or C-terminus of proteins which can be used in methods of purifying a protein from cells. Cells expressing a peptide comprising an affinity tag can be expressed with a signal sequence in the supernatant/cell culture medium. Cells expressing a peptide comprising an affinity tag can also be pelleted, lysed, and the cell lysate applied to a column, resin or other solid support that displays a ligand to the affinity tags. The affinity tag and any fused peptides are bound to the solid support, which can also be washed several times with buffer to eliminate unbound (contaminant) proteins. A protein of interest, if attached to an affinity tag, can be eluted from the solid support via a buffer that causes the affinity tag to dissociate from the ligand resulting in a purified protein, or can be cleaved from the bound affinity tag using a soluble protease. As disclosed herein, the affinity tag is cleaved through the self-cleaving mechanism of the C-intein segment in the active intein complex.

[0086] Affinity tags can also be used to facilitate the purification of a protein of interest using the disclosed modified peptides through a variety of methods, including, but not limited to, selective precipitation, ion exchange chromatography, binding to precipitation-capable ligands, dialysis (by changing the size and/or charge of the target protein) and other highly selective separation methods.

[0087] In some aspects, affinity tags can be used that do not actually bind to a ligand, but instead either selectively precipitate or act as ligands for immobilized corresponding binding domains. In these instances, the tags are more generally referred to as purification tags. For example, the ELP tag selectively precipitates under specific salt and temperature conditions, allowing fused peptides to be purified by centrifugation. Another example is the antibody Fc domain, which serves as a ligand for immobilized protein A or Protein G-binding domains.

Proteins of Interest

[0088] Target proteins for all protocols are: any recombinant proteins, especially proteins requiring native or near native N-terminal sequences, for example therapeutic protein candidates, biologics, antibody fragments, antibody mimetics, protein scaffolds, enzymes, recombinant proteins or peptides, such as growth factors, cytokines, chemokines, hormones, antigen (viral, bacterial, yeast, mammalian) production, vaccine production, cell surface receptors, fusion proteins.

[0089] The invention will now be described more closely in association with some non-limiting examples and the accompanying drawings.

Experimental Part

[0090] The invention will be described more closely in association with some non-limiting examples and the accompanying drawings.

[0091] In the present invention the following 5 constructs were evaluated:

TABLE 1

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A52      ALSYETEILTVEYGLLPIGKIVEKRIECTVYSVDNNGNIYTQPVAQWHRDGEQVEFYEYCL
A53      ALSYDTEILTVEYGLPLPGKIVEENIECTVYSVDKNGFVYTQP IAQWHRNDRGEQVEFYEYDL
B97_K24E/R25N  ALSYETEILTVEYGLLPIGKIVEENIECTVYSVDNNGNIYTQPVAQWHRDGEQVEFYEYCL
B82_K24E      ALSYETEILTVEYGLLPIGKIVEERIECTVYSVDNNGNIYTQPVAQWHRDGEQVEFYEYCL
B83_R25N      ALSYETEILTVEYGLLPIGKIVEKNIECTVYSVDNNGNIYTQPVAQWHRDGEQVEFYEYCL
          ****:*****:*****: .*****: *  : ***:****:***** *

A52      EDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRVGGSGDYKDDDDKGGSGHHHHHH
A53      EDGSIIRATKDHKFMTVDGEMLPIDEIFEQGLDLKQVGGSGDYKDDDDKGGSGHHHHHH
B97_K24E/R25N  EDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRVGGSGDYKDDDDKGGSGHHHHHH
B82_K24E      EDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRVGGSGDYKDDDDKGGSGHHHHHH
B83_R25N      EDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRVGGSGDYKDDDDKGGSGHHHHHH
          ****:*****. ** :*****: *** .*****

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Non-Intein sequences

[0085] Examples of affinity include, but are not limited to, maltose binding protein, which can bind to immobilized maltose to facilitate purification of the fused target protein; Chitin binding protein, which can bind to immobilized chitin; Glutathione S transferase, which can bind to immobilized glutathione; poly-histidine, which can bind to immobilized chelated metals; FLAG octapeptide, which can bind to immobilized anti-FLAG antibodies.

Construct Cultivation and Expression:

[0092] Start cultures were diluted 1:100 in 100 mL LB+neo in shake flasks (done in triplicates for all 5 constructs) and incubated at 37 until OD600 was 0.6-1. Once target OD was reached, the flasks were transferred to a cooled incubator (22 degrees) and 0.5 mM IPTG was added for induction over night (exactly 18 h). Induced cultivations

were pelleted sequentially at 5000 g (+7 degrees) in 50 mL falcon tubes and pellets were weighed.

Solubility Testing

- [0093] Weighed pellets (ca 0.8 grams) were resuspended in 20 volumes of 1× PBS. For each of the constructs;
- [0094] 1. 20 μL was saved for SDS-PAGE (Whole cell Lysate/WCL)
- [0095] 2. 5 mL was added to 50 mL falcon tube and
- [0096] a. centrifuged for 20 min at 5000 g (6 degrees)
- [0097] b. pellet was resuspended in 5 mL 8M Urea
- [0098] c. incubated by end-to-end for ~1 h at room temp
- [0099] d. Centrifuged at 20000 g for 20 min at 6 degrees
- [0100] e. 20 μL of supernatant was saved for SDS-PAGE (Urea)
- [0101] f. Rest of supernatant was used for Biacore CFCA analysis
- [0102] 3. 10 mL was added to to a 50 mL falcon tube and
- [0103] a. Sonicated on ice for 1 min on time (2 sec on/4 sec off pulses) at 30% amplitude
- [0104] b. Centrifuged at 20000 g for 20 min at 6 degrees
- [0105] c. 20 μL of supernatant was saved for SDS-PAGE (Sonic.)
- [0106] d. Rest of supernatant was used for Biacore CFCA analysis
- [0107] 4. 500 μL was added to a 1.5 mL tube and
- [0108] a. Centrifuged for 20 min at 5000 g at 6 degrees
- [0109] b. Resuspended pellet in 500 μL 1% NP40 buffer
- [0110] c. Incubated for 40 min with 1200 rpm skaking (Mathilda block) at room temperature
- [0111] d. Centrifuged for 20 min at 12000 g at 6 degrees
- [0112] e. Saved ca 250 μL of the supernatant
- [0113] f. 20 μL of supernatant was saved for SDS-PAGE (NP40)
- [0114] g. Rest of supernatant was used for BiaCore CFCA analysis
- [0115] To the 20 μL samples 40 μL SDS-PAGE buffer was added and boiled at 95 degrees for 5 min prior to loading the samples to a 15% gel.

Solubility was Evaluated by SDS-PAGE Analysis Samples and Calculated Accordingly:

[0116]

$$\frac{\text{Extraction method densitometric signal/Whole cell lysate (WCL) densitometric signal} \times 100\%}{}$$

- [0117] FIG. 1 shows SDS-PAGE analysis of representative supernatants after using different extraction techniques. 20 μL of supernatants were mixed with 40 μL of 2× Laemmli sample buffer and boiled for 5 minutes at 95 degrees celsius prior to loading on a 15% homogenous SDS-PAGE gel. Gel was electrophoresed for 1 h and 50 min at 600V and stained by coomassie for approximately two hours. After extensive destaining, gel was imaged using an Amersham AI600 imager.
- [0118] FIG. 2 Shows solubility determined after densitometric evaluation of SDS-PAGE analysis. Extracts from

three different cell-cultures for each construct were analysed and ligand band densitometry was measured using ImageQuant TL software. Solubility was calculated based on the following formula:

$$\frac{\text{Densitometry(extraction method X)/Densitometry (WCL)} \times 100\% = \% \text{ solubility}}$$

[0119] Bars show the average solubility of different extraction methods compared to whole cell lysate and the error bars show the standard deviation. All constructs showed significantly better solubility than A52 for the sonicated samples analysed. A53, B97 and B82 showed significantly higher solubility as compared to A52 samples. Statistical significance was determined by one-way ANOVA with Dunnet's post test using A52 as control sample. *p-value <0.05, ** p-value <0.01, *** p-value <0.001, **** p-value <0.0001.

Construct name	NP40, mean solubility	Sonication, mean solubility
A52	7.4%	22.1%
A53	67.6%	104.7%
B97	37.3%	92.8%
B82	44.3%	62.8%
B83	12.4%	36.6%

[0120] Urea extraction method showed no significant difference as compared to WCL and independent of the construct analysed.

Biacore CFCA Analysis

- [0121] Determination of the soluble N-intein ratio of the various protein extracts was further analyzed by SPR binding analysis using a FLAG-epitope (DYKDDDDK) as a detection-tag at the C-terminus of the constructs. Calibration-free concentration analysis, (CFCA), was done in a Biacore T200 instrument using a mouse monoclonal ANTI-FLAG M2 antibody. Sensor chips, CM5 series S were immobilized with the anti-FLAG antibody using an amine coupling kit. 10 mM sodium acetate pH 4.0 was used as immobilization buffer, HBS-EP+pH 7.4 as a running buffer and Glycine-HCl pH 2.5 as regeneration buffer. The immobilization levels were about 8000-10000 RU. Supernatant samples from the different extractions described above, were diluted from 150-5000 times in HBS-EP+running buffer before anlysis by the CFCA method. The molecular weights of the different protein constructs ranged from 13.5-13.6 kDa and this was used to calculate the diffusion coefficient at 20° C., (1.13389E-10 m²/s). The default Biacore method for CFCA was used as a starting point for setting up the final method. Sample concentrations were determined by using the Biacore T200 evaluation software.
- [0122] FIG. 3 shows N-intein concentrations in supernatants from different extracts determined by Biacore CFCA analysis. Extracts from three different cell-cultures for each construct were analysed. Bars show the average concentration and the error bars show the standard deviation.
- [0123] N-intein concentration in the supernatants after extraction and clarification using different extraction methods is used for the calculation of soluble N-intein ratios. The NP40 detergent buffer causes a mild release of soluble proteins from the cells. Ultra-sonication is a mechanical extraction technique causing vigorous cell disruption, releasing soluble proteins. Urea at high concentration is a

denaturing extraction method causing the release of both soluble proteins and insoluble proteins found in inclusion bodies from the cells. Boiling of the cell pellets in a SDS sample buffer causes complete solubilization of both soluble and insoluble N-intein and is used as the reference for total amount of expressed N-intein. A high N-intein concentration in supernatants after extraction with non-denaturing extraction methods, (NP40 and sonication) compared with denaturing methods, (Urea and SDS) indicate a high solubility. The CFCA analysis show that the A53 and B97 constructs have a high solubility whereas A52 has a very poor solubility, FIG. 4.

[0124] Statistical analysis show that the modified constructs B82, B83 and B97 are significantly more soluble compared with the non-modified A52 construct when using mild non-denaturing extraction methods.

Solubility Evaluated by SPR Binding Analysis is Calculated Accordingly:

[0125]

$$\frac{\text{Extraction method N-intein concentration/SDS}}{\text{extracted N-intein concentration}} \times 100\%$$

Results

[0126] Sodium dodecyl sulfate, SDS, is an ionic detergent that binds to proteins through ionic and hydrophobic interactions and solubilizes proteins by altering their secondary and tertiary structure. SDS is routinely used in polyacrylamide gel electrophoresis, (SDS-PAGE) to separate, characterize and quantify proteins. SDS has been used in these example experiments as a universal protein solubilizing reagent used for quantification of the total amount of protein in different extracts, both soluble and insoluble for subsequent separation, detection and quantification by densitometric analysis of SDS-PAGE gels and Biacore calibration free concentration analysis, CFCA. The concentration of different constructs in SDS solubilized sample extracts is normalized to 100% for comparison with the concentration of the different protein constructs derived in the supernatants after centrifugal clarification of extracts using different methods.

[0127] A mild method for extracting soluble proteins only is the use of a non-ionic detergent NP40. NP40 at 1% (w/v) is added to a Tris-HCl buffer, pH 7.5 containing 150 mM sodium chloride and is simply used by resuspending harvested bacterial cell pellets followed by mixing during 1 hour. After incubation the cell suspension is clarified by centrifugation to remove insoluble material.

[0128] Ultra-sonication or sonication, is an extraction method for proteins that uses mechanical energy from a probe to disintegrate cells for the release of soluble cell components. Cells are resuspended in a non-denaturing buffer like phosphate buffered saline, PBS at pH 7.4 to control the pH during the release of cellular components. Sonication is a very efficient and reliable tool for cell disintegration that allows for a complete control over the sonication parameters. This ensures a high selectivity on materials release and product purity. After sonication, the lysate is clarified by supernatant and the insoluble pellet is removed.

[0129] Chaotropic salts like Urea can be used for the release of both soluble and insoluble proteins from cells. Urea is compatible with a wide range range of analytical

methods in contrast to SDS detergent that is more likely to interfere with some commonly used analytical methods. Urea is commonly used at 8 M to ensure maximum denaturing conditions and can be dissolved in water. Cells are resuspended in the Urea solution followed by mixing during 1 hour. The extract is then clarified by centrifugation to remove the insoluble pellet.

[0130] SDS denatures proteins when heated and imparts a strong negative charge to all proteins. SDS binds strongly to proteins in the ratio of one SDS molecule per two amino acids. This makes SDS extraction a very efficient method to assess the amount of total protein, both soluble and insoluble. In general, a 2% (w/v) SDS concentration in a buffer solution between pH 6.7-7.5 is added to an equal volume of cell suspension from a cell harvest followed by mixing and heating at 95° C. for 5 minutes. Then the samples are cooled down to room temperature before centrifugation and analysis.

[0131] FIG. 3. shows the concentration of different N-intein constructs in the supernatants after extraction of proteins in the cell harvest by the use of different methods. The amount of cells and the extraction volumes were normalized prior to extraction so that the actual concentration can be directly compared. Each bar shows the average concentration for a certain construct derived from the extraction of cells from three different cell cultures. The error bars show the standard deviation. As can be seen in FIG. 3. the concentration of the protein constructs are highest in the supernatants after extraction using SDS and Urea. The relative difference between the concentration of the different constructs reflect a varying degree of expression from the different cell cultures. Constructs A52 and B97 had the highest N-intein expression in total according to concentration in the SDS extracts, 871 and 803 µg/ml respectively. N-intein concentrations from the Urea extracts are generally lower compared with SDS extracts but follows roughly the same pattern. The interesting findings can be seen in the N-intein concentrations from the sonication and NP40 extracts where only the soluble proteins are found. A52, a construct that does not comprise the substitution mutations K24E or R25N has the lowest concentration of N-intein compared with the other constructs with 27.5 µg/ml in NP40 extracts and 37.9 µg/ml in sonicated samples. The construct B97 comprising the K24E and R25N substitutions has a relatively high concentration of soluble N-intein in NP40 extracts, 180.3 µg/ml and in sonicated extracts, 662.3 µg/ml. This difference is more pronounced in FIG. 4., where the N-intein concentration for each respective construct and extraction method is compared with the N-intein concentration after SDS extraction of each respective construct. SDS bars are omitted since they all give the ratio 1, equal to 100%. The construct A52 lacking the mutations at position 24 and 25 has only 3 and 4% N-intein in extracts from NP40 and sonication respectively compared with SDS extracts. A single substitution, R25N in construct B83 results in a higher ratio relative to SDS extracts, 11% and 25% respectively for NP40 and sonication extracts. A single substitution, K24E in construct B82 results in a higher ratio relative to SDS extracts, 19% and 49% respectively for NP40 and sonication extracts. Construct B97, with two amino acid substitutions at position 24 and 25, K24E and R25N, results in a higher ratio relative to SDS extracts, 22% and 82% respectively for NP40 and sonication extracts.

[0132] In summary, the solubility ranges achieved according to the invention in the above experiments are:

[0133] At least 10-40% soluble N-intein with a single-point mutation of R at position 25, preferred N or non-positive amino acid. At least 46-52% soluble N-intein with a single-point mutation of K at position 24, preferred E or non-positive amino acid. At least 76-88% soluble N-intein with mutations at positions 24 and 25, preferred K24E and R25N or non-positive amino acids. These values are based on Biacore CFCA data on sonicated samples.

RATKDHKFMVTVDGQMLPIDEIFERELDLMRV

(SEQ ID NO 1), wherein the protein variant comprises an amino acid substitution from K to E in position 24 of SEQ ID NO 1 and R to N in position 25 of SEQ ID NO 1 to increase solubility in aqueous buffers, and wherein optionally one or more C is/are mutated to non-Cystein residues, preferably S or A.

6. The N-intein protein variant of claim 1, wherein the solubility in aqueous buffer is at least 10-40% soluble N-intein with a single-point mutation of R at position 25,

SEQUENCE LISTING

<160> NUMBER OF SEQ ID NOS: 1

<210> SEQ ID NO 1

<211> LENGTH: 97

<212> TYPE: PRT

<213> ORGANISM: *Nostoc punctiforme*

<400> SEQUENCE: 1

Cys Leu Ser Tyr Glu Thr Glu Ile Leu Thr Val Glu Tyr Gly Leu Leu
1 5 10 15

Pro Ile Gly Lys Ile Val Glu Lys Arg Ile Glu Cys Thr Val Tyr Ser
20 25 30

Val Asp Asn Asn Gly Asn Ile Tyr Thr Gln Pro Val Ala Gln Trp His
35 40 45

Asp Arg Gly Glu Gln Glu Val Phe Glu Tyr Cys Leu Glu Asp Gly Ser
50 55 60

Leu Ile Arg Ala Thr Lys Asp His Lys Phe Met Thr Val Asp Gly Gln
65 70 75 80

Met Leu Pro Ile Asp Glu Ile Phe Glu Arg Glu Leu Asp Leu Met Arg
85 90 95

Val

1. An N-intein protein variant derived from wildtype *Nostoc punctiforme* (Npu) or sequences having at least 95% homology therewith comprising at least one amino acid substitution of a native split intein, wherein the N-intein protein variant sequence includes a mutation in at least position 24 and/or position 25 as measured from the initial catalytic cysteine and wherein the substituted amino acid provides increased solubility in aqueous buffers compared to the native N-intein protein sequence or a consensus N-intein sequence.

2. The N-intein protein variant of claim 1 wherein the substituted amino acid(s) that provide increased solubility is a non-positive amino acid.

3. The N-intein protein variant of claim 1, wherein the substituted amino acid that provide increased solubility is K24E.

4. The N-intein protein variant of claim 1, wherein the substituted amino acid that provide increased solubility is R25N.

5. An N-intein protein variant of the wildtype N-intein domain of *Nostoc punctiforme* (Npu) wherein the wildtype Npu N-intein domain comprises the following sequence:

CLSYETEILTVEYGLLPIG-
KIVEKRIECTVYSVDNNG-
NIYTQPAQWHDGRGEQEVFE YCLEDGSLI-

preferred N or non-positive amino acid; at least 46-52% soluble N-intein with a single-point mutation of K at position 24, preferred E or non-positive amino acid; and at least 76-88% soluble N-intein with mutations at positions 24 and 25, preferred K24E and R25N or non-positive amino acids.

7. The N-intein protein variant according to claim 1, which is attached to a solid phase, such as a membrane, fiber, particle, bead or chip.

8. The N-intein protein variant sequence according to claim 7, wherein the solid phased is a chromatography resin of natural or synthetic origin.

9. The N-intein protein variant according to claim 7, wherein the solid phase is a chromatography resin, such as a natural or synthetic resin, preferably a polysaccharide such as agarose.

10. The N-intein protein variant according to claim 9, wherein the solid phase is provided with embedded magnetic particles.

11. The N-intein protein variant according to claim 9, wherein the solid phase is a non-diffusion limited resin/fibrous material.

12. The N-intein protein variant according to claim 1, wherein the N-intein is coupled to the solid phase via a Lys-tail, comprising one or more Lys, on the C-terminal.

13. The N-intein protein variant according to claim **1**, wherein the N-intein is coupled to the solid phase via a Cys-tail on the C-terminal.

14. The N-intein protein variant according to claim **1**, wherein 0.2-2 $\mu\text{mole/ml}$ N-intein is coupled per ml solid phase, preferably chromatography resin (ml swollen gel).

15. A split intein system comprising a N-intein protein variant according to claim **1**, attached to a solid phase, and a C-intein sequence which is co-expressed with a POI (protein of interest), wherein the C-intein acts as a tag on the POI and the expressed C-intein binds to said N-intein protein variant.

16. Split intein system according to claim **15**, wherein the C-intein sequence is a native split intein C-intein sequence or engineered variants thereof.

17. Split intein system according to claim **15**, wherein the POI's are: proteins requiring native or near native N-terminal sequences, for example therapeutic protein candidates, biologics, antibody fragments, antibody mimetics, enzymes, recombinant proteins or peptides, such as growth factors, cytokines, chemokines, hormones, antigen (viral, bacterial, yeast, mammalian) production, vaccine production, cell surface receptors, fusion proteins.

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